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Menna et al.

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(54) **STRETCHER**

USPC 296/20, 26.1; 5/611, 86.1
See application file for complete search history.

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A61G 1/02 (2006.01)

A61G 3/02 (2006.01)

(52) **U.S. Cl.**

CPC **A61G 1/0562** (2013.01); **A61G 1/0212** (2013.01); **A61G 3/0245** (2013.01); **A61G 3/0254** (2013.01); **A61G 3/0272** (2013.01)

(58) **Field of Classification Search**

CPC .. A61G 1/0562; A61G 1/0212; A61G 3/0245; A61G 3/0254; A61G 3/0272; A61G 2203/30; A61G 2203/42; A61G 1/0237; A61G 3/0875; A61G 1/0262; A61G 1/0256; A61G 7/012; A61G 1/04; A61G 7/018; A61G 1/0293; A61G 1/0567; A61G 1/048; A61G 1/0243; A61G 2203/32; A61G 2203/40; A61G 1/013; A61G 1/0275; A61G 2203/726

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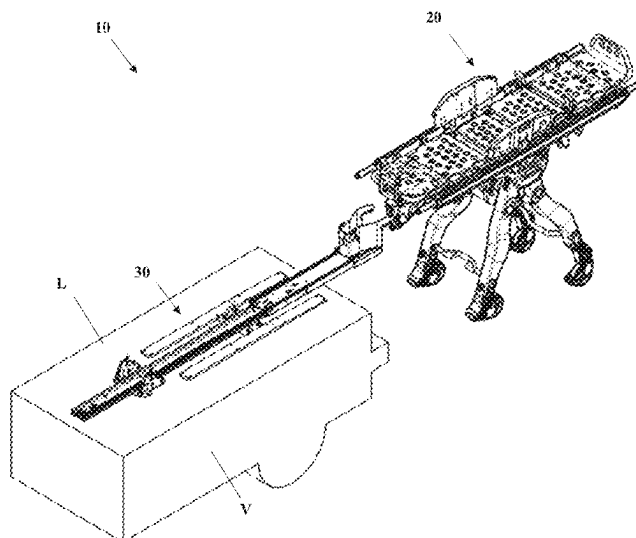
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(57) **ABSTRACT**

A stretcher including a support frame for supporting a patient, front legs and rear legs rotatably coupled to the support frame around a respective first and second rotation axis. Each front and rear leg supports respectively a front wheel holder frame hinged to the respective front leg around a first oscillation axis parallel to the first rotation axis and supporting a front wheel resting on a rest plane and a rear wheel holder frame hinged to the respective rear leg around a second oscillation axis parallel to the second rotation axis and supporting a rear wheel resting on a rest plane. The stretcher has a handling arrangement for the front wheel holder frame and the rear wheel holder frame, and a control module configured to keep the first and second pivot axes always orthogonal to the rest plane on which the front and rear wheels roll.

8 Claims, 16 Drawing Sheets



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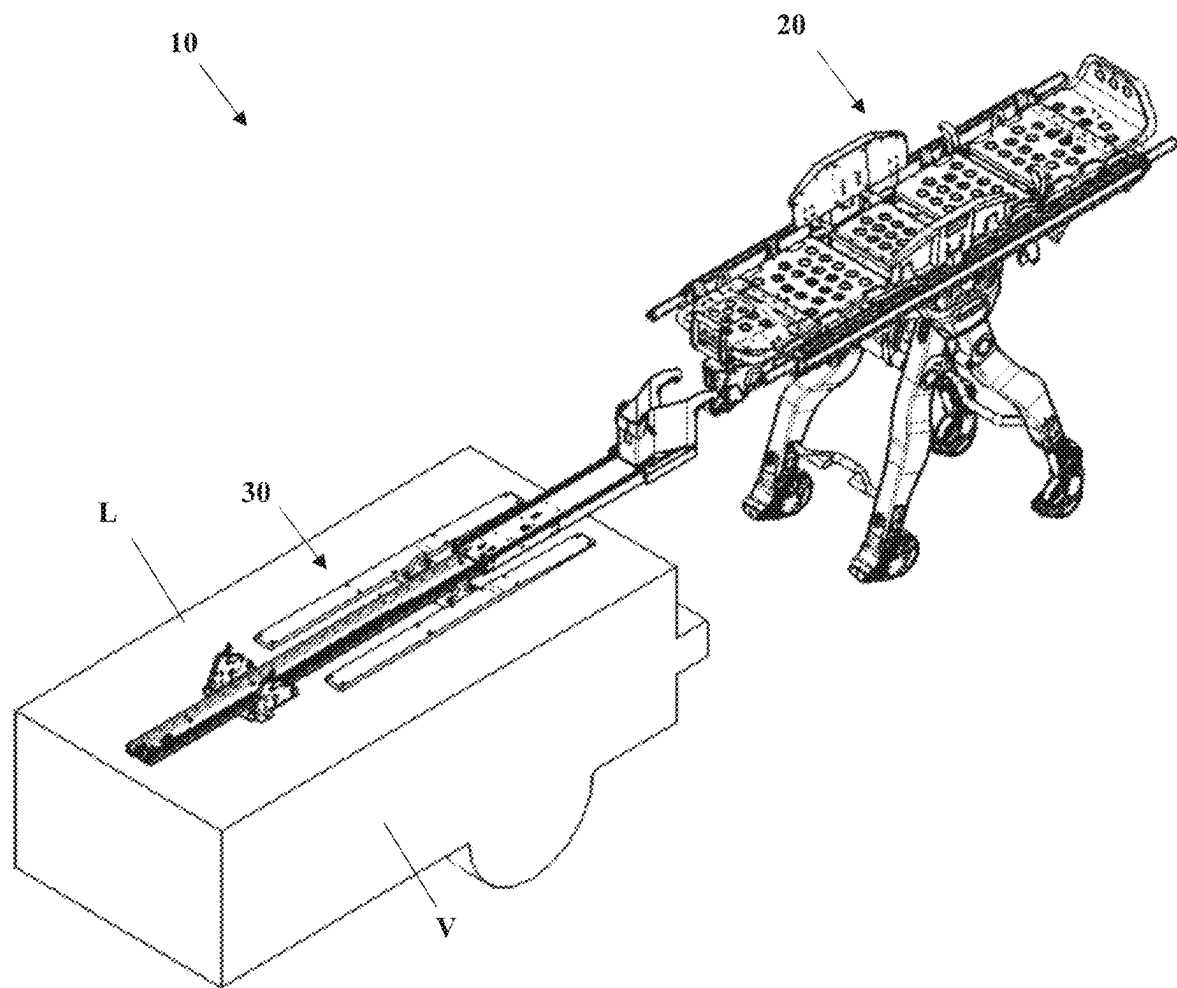


FIG. 1

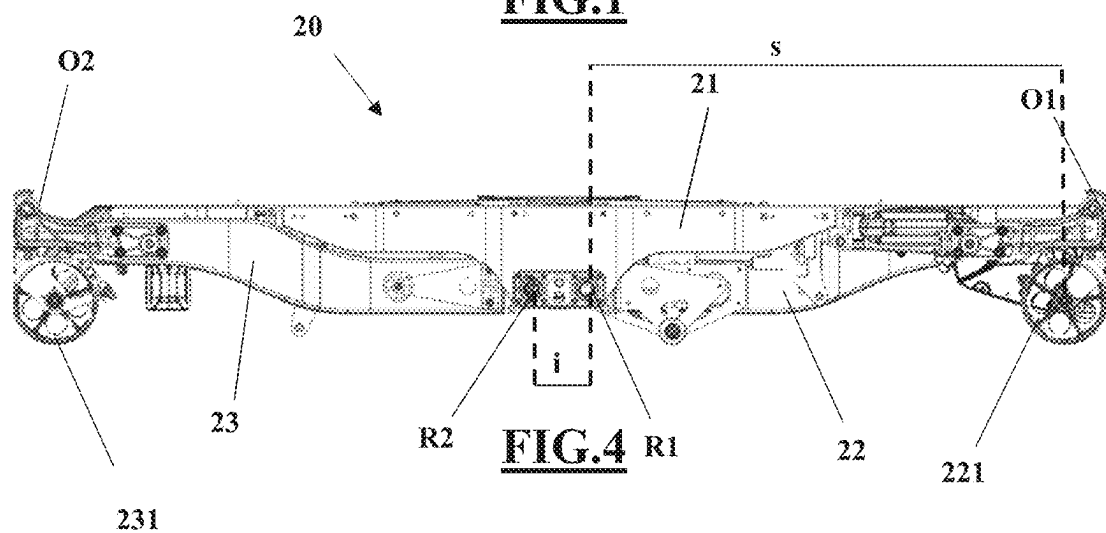


FIG. 4

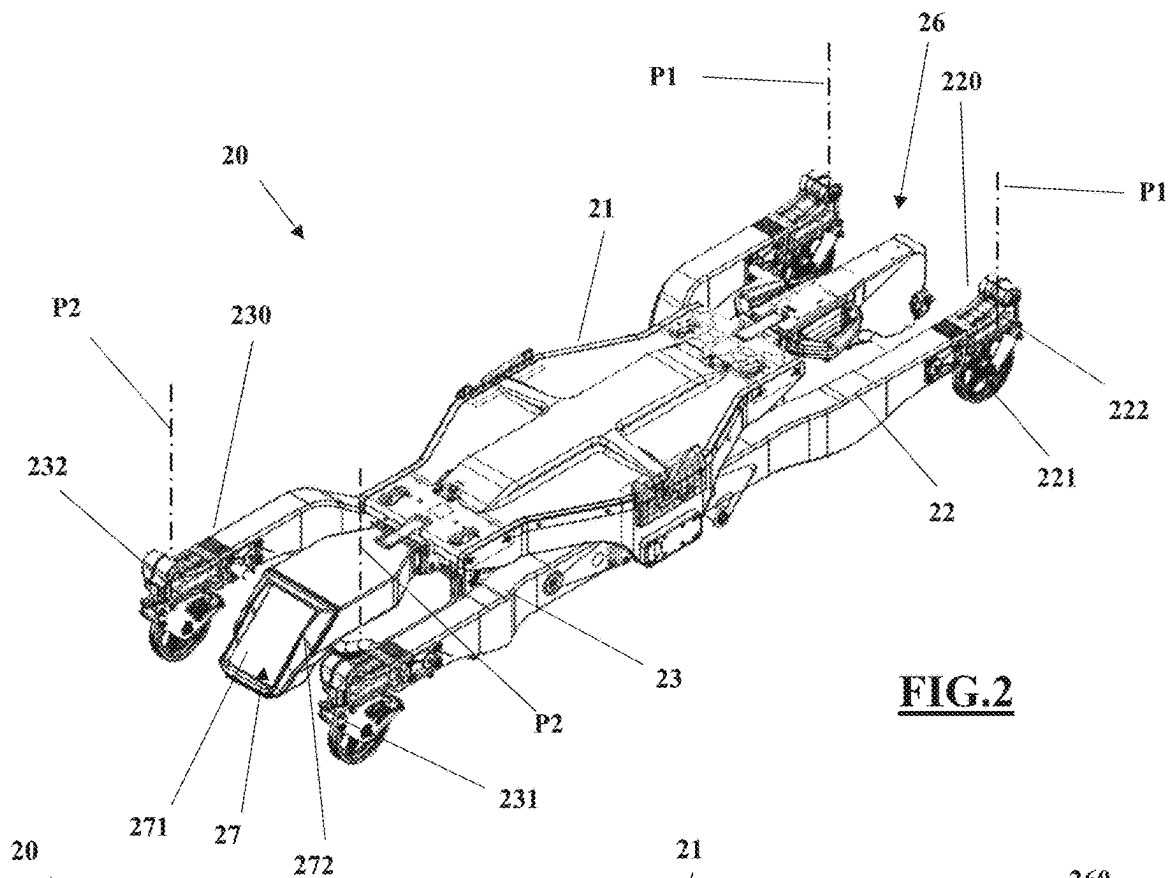


FIG. 2

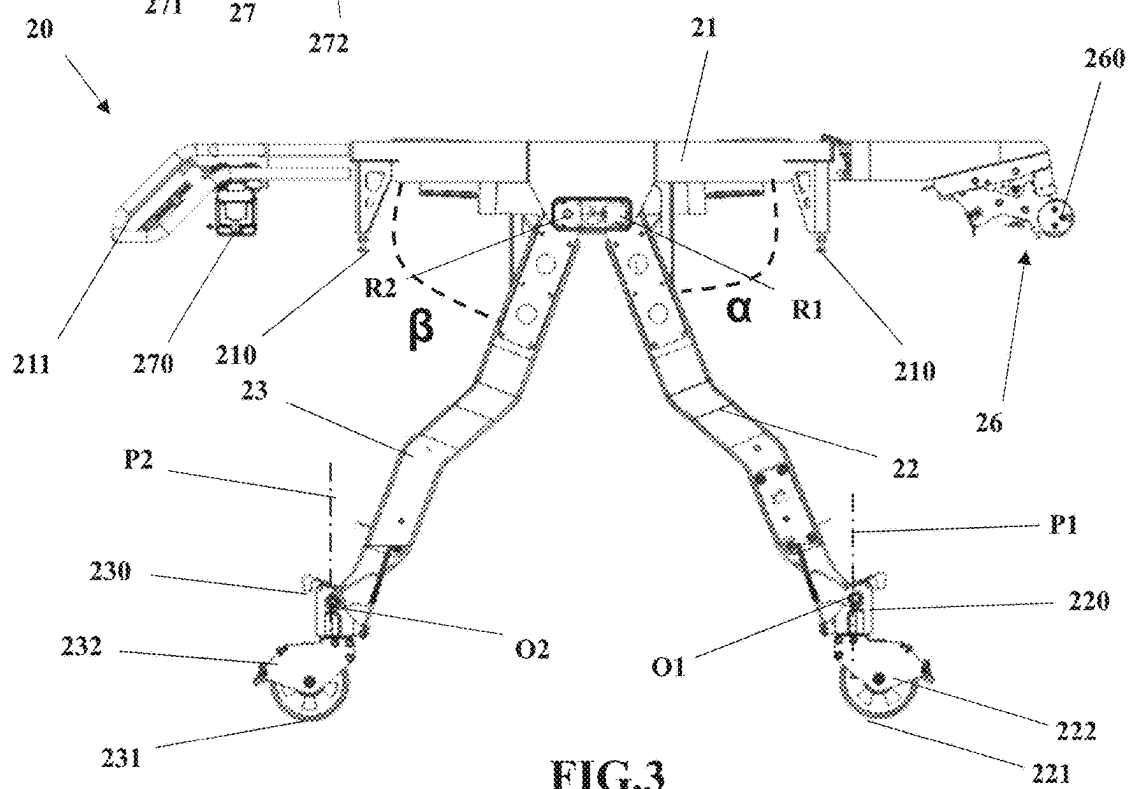


FIG. 3

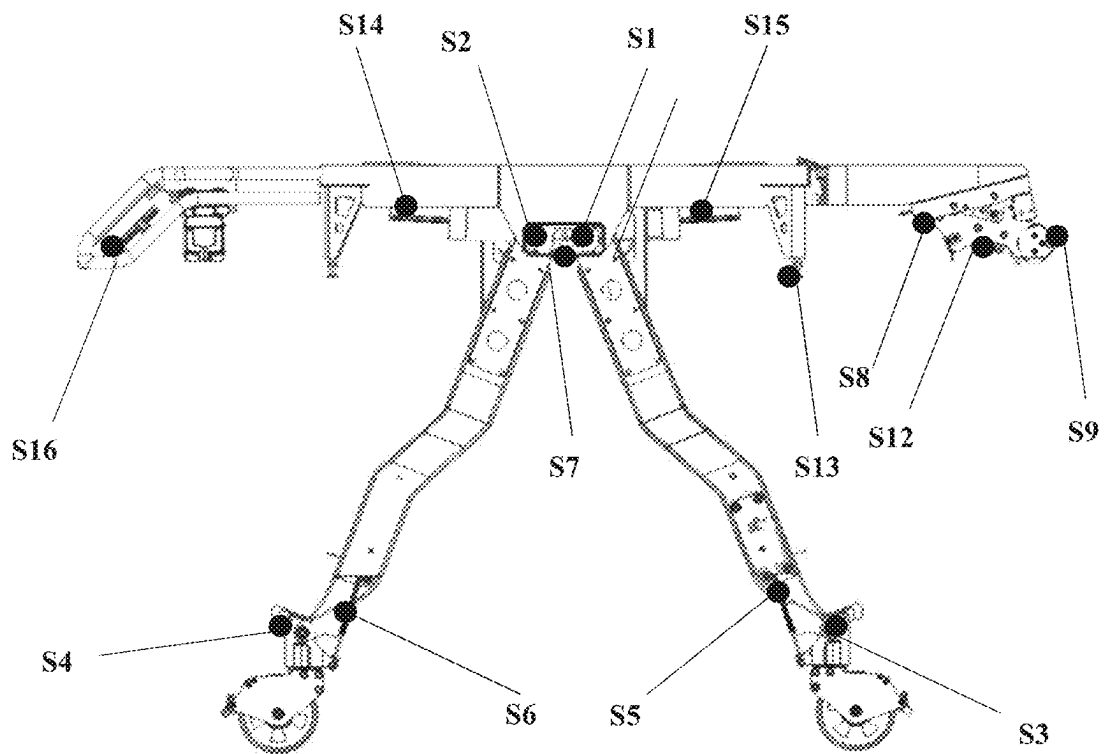


FIG. 5

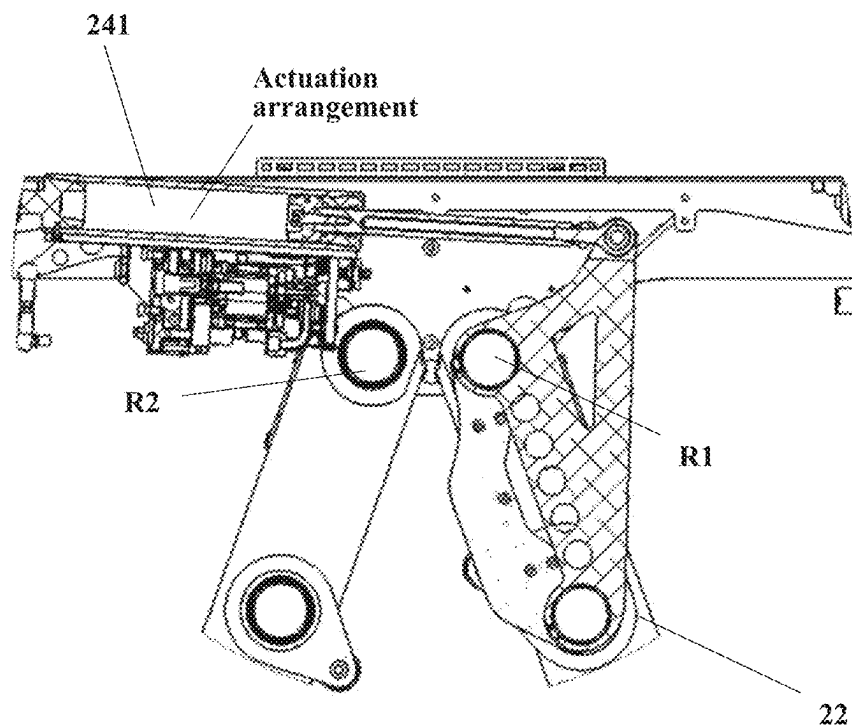


FIG. 6A

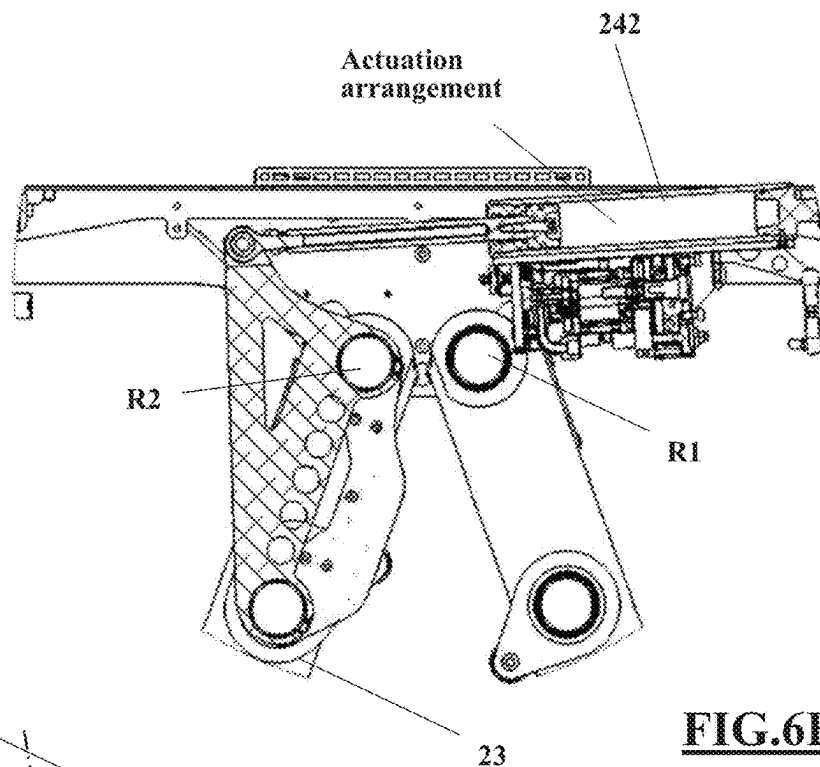


FIG. 6B

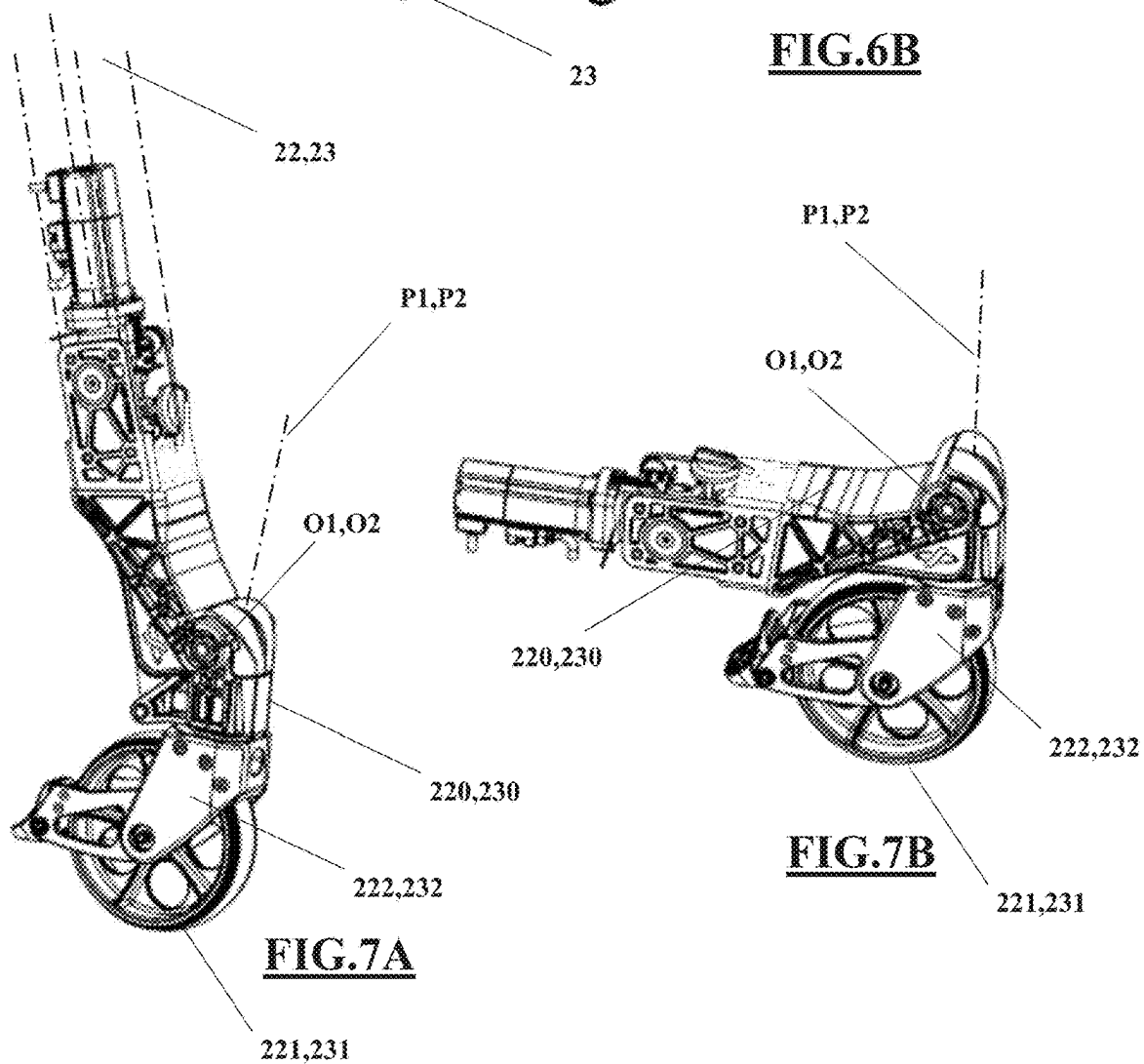


FIG. 7A

FIG. 7B

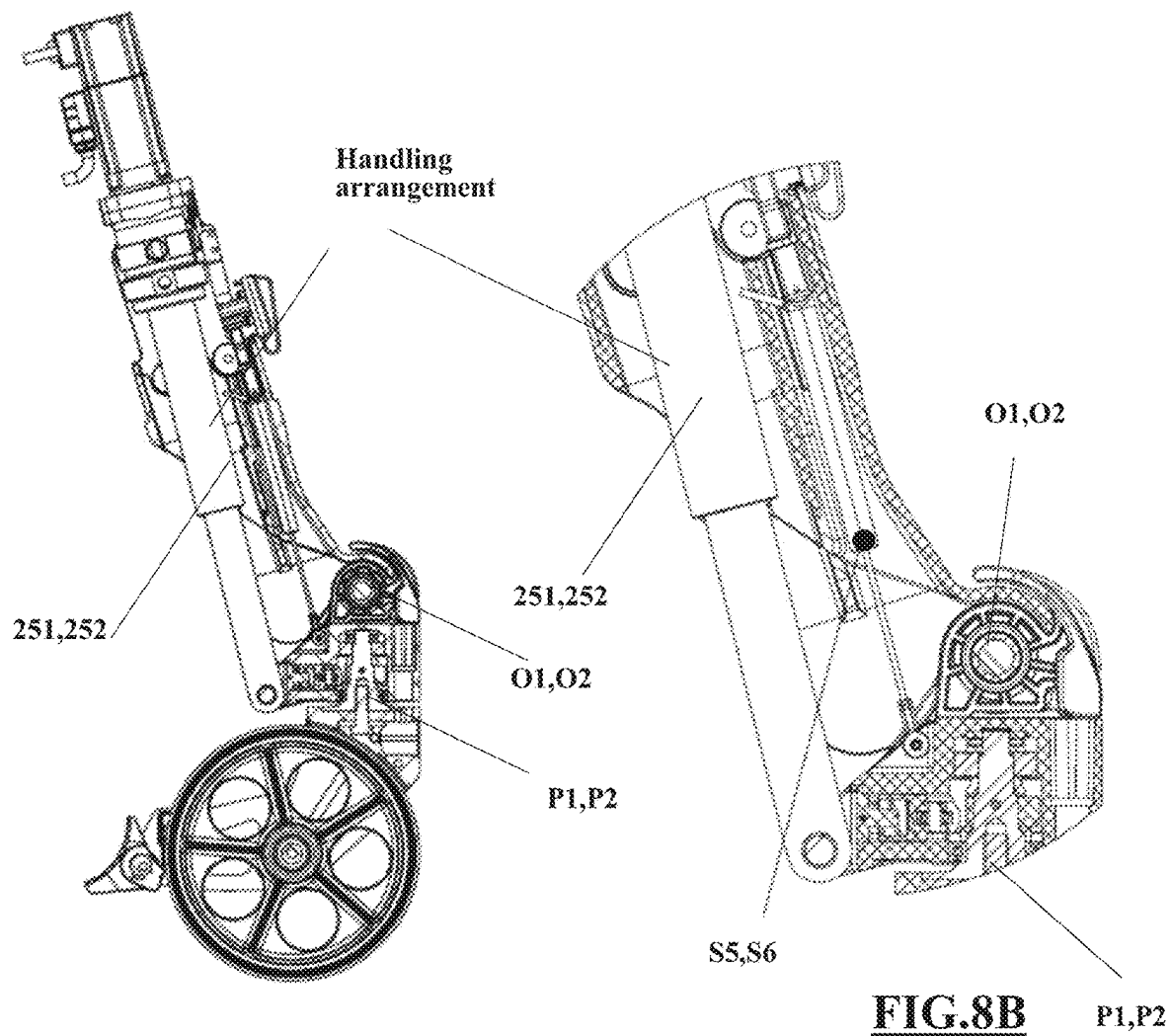


FIG.8A

FIG.8B

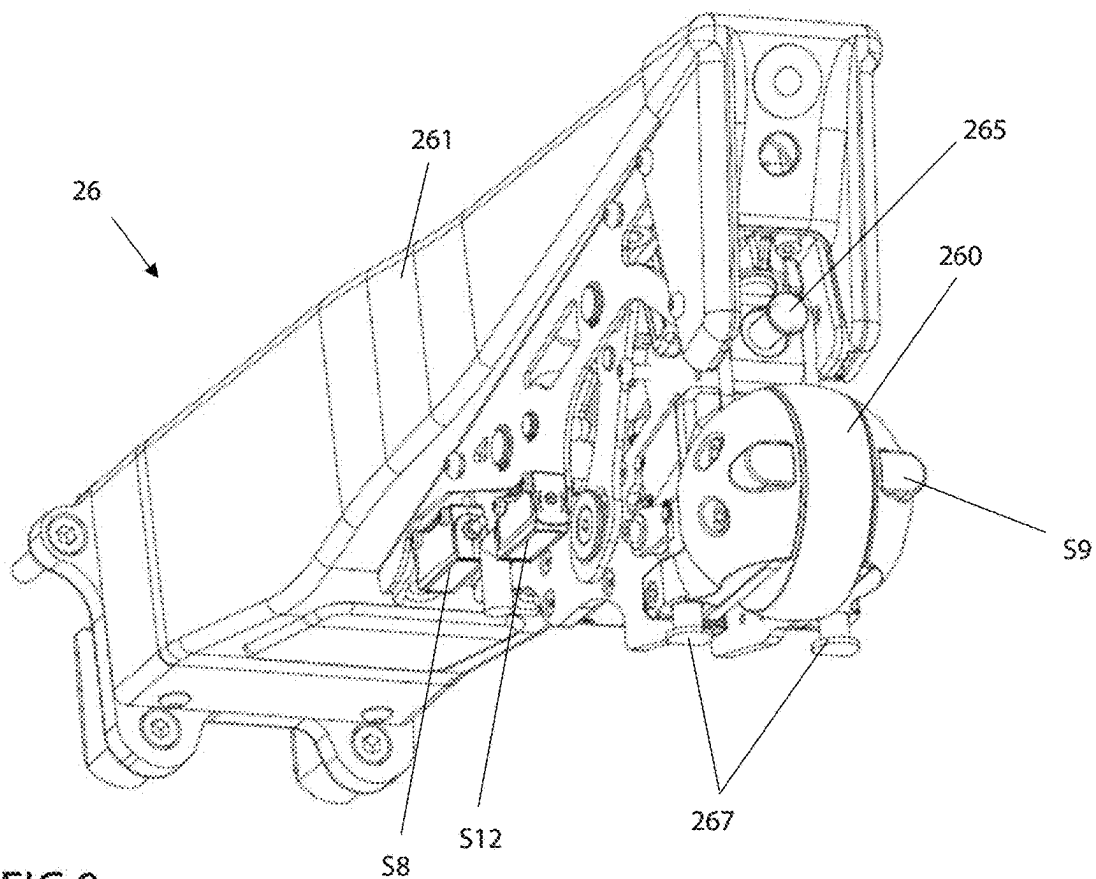


FIG. 9

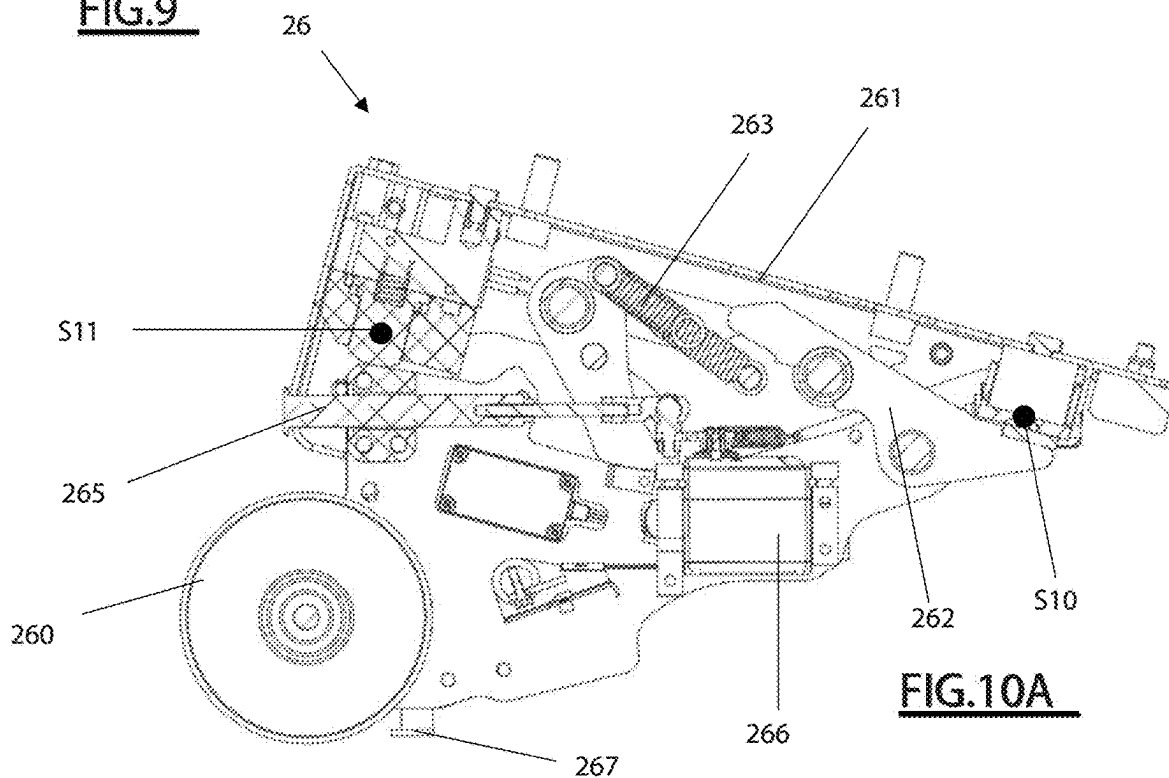
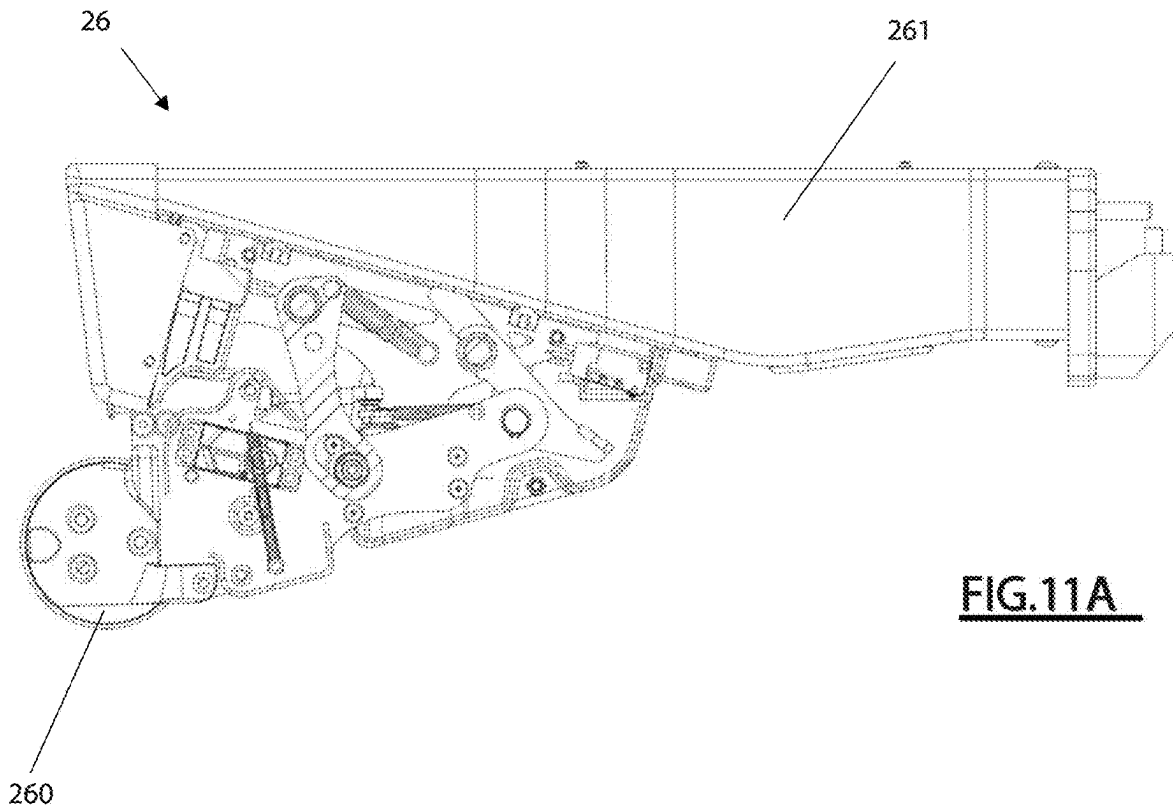
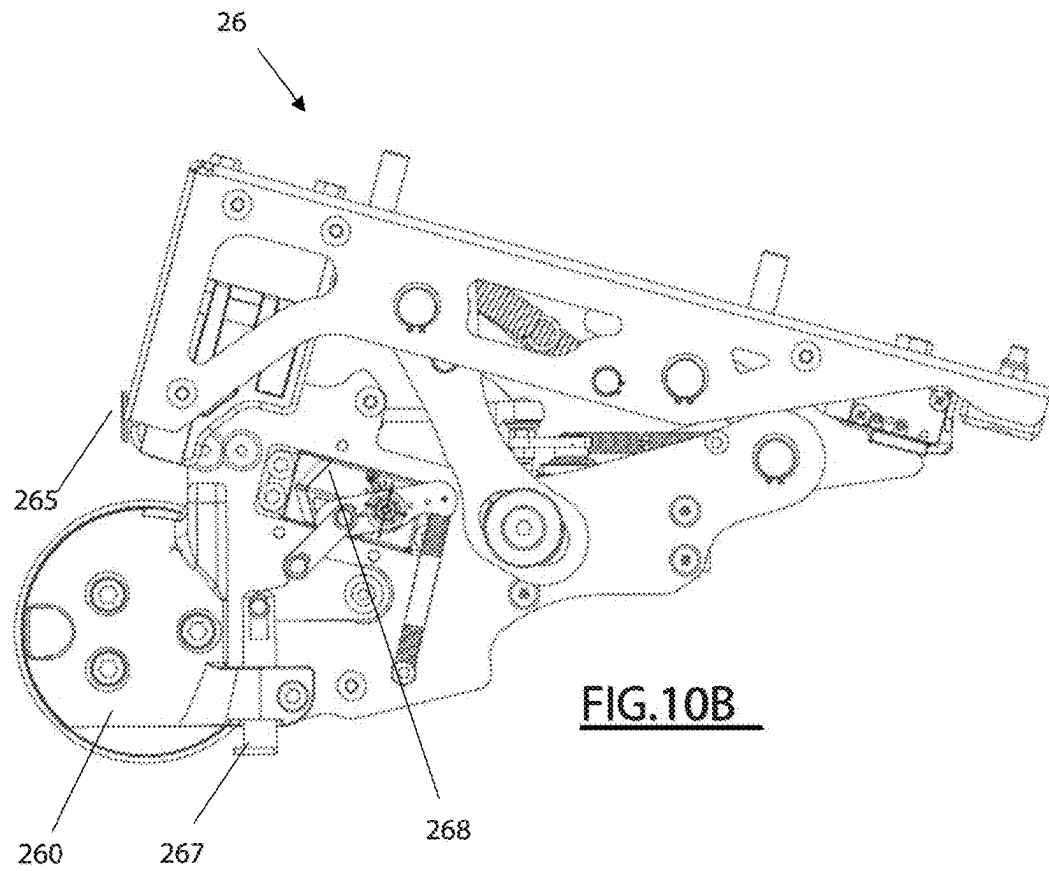
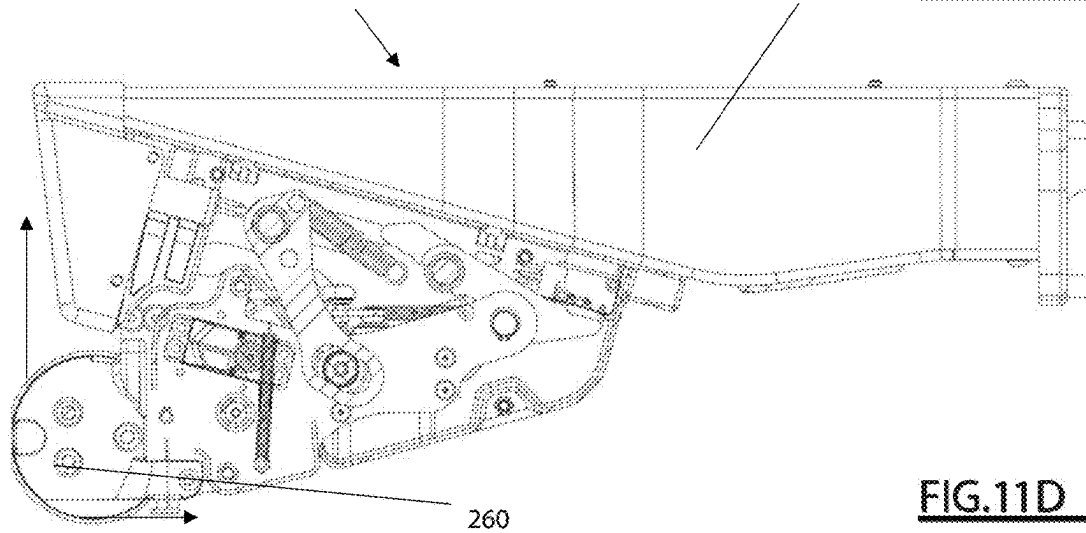
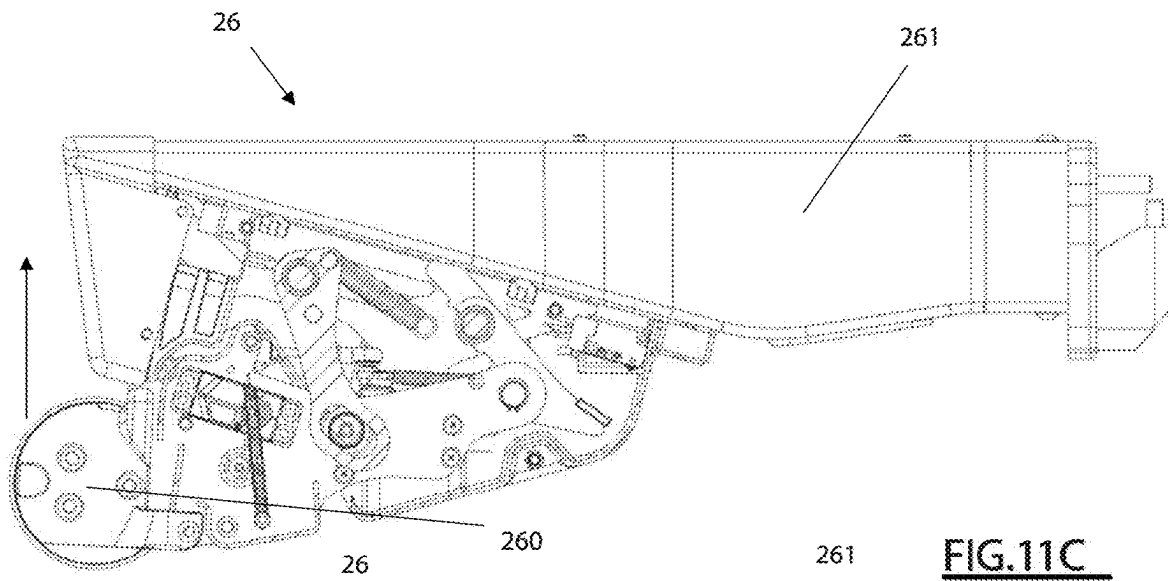
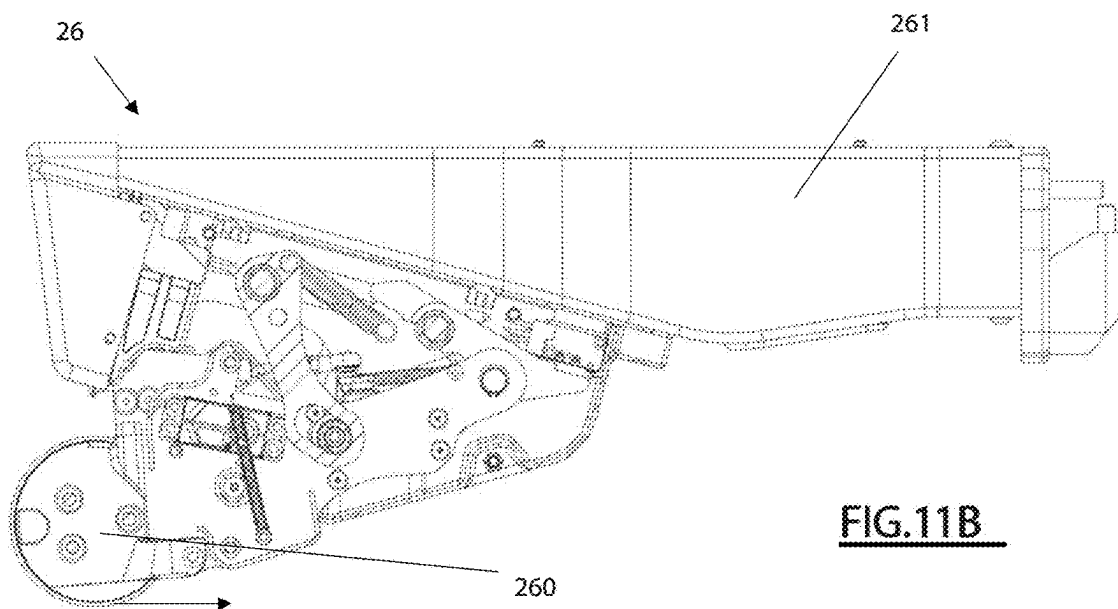
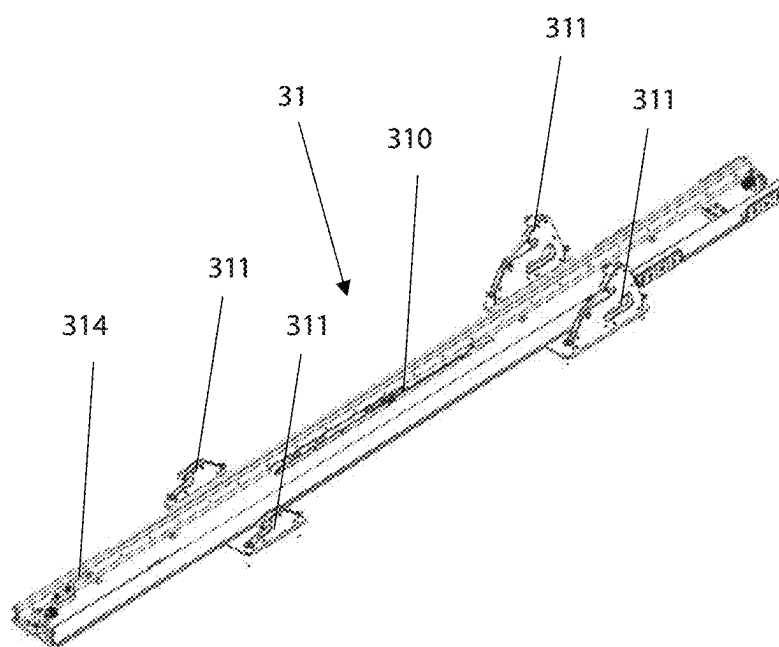
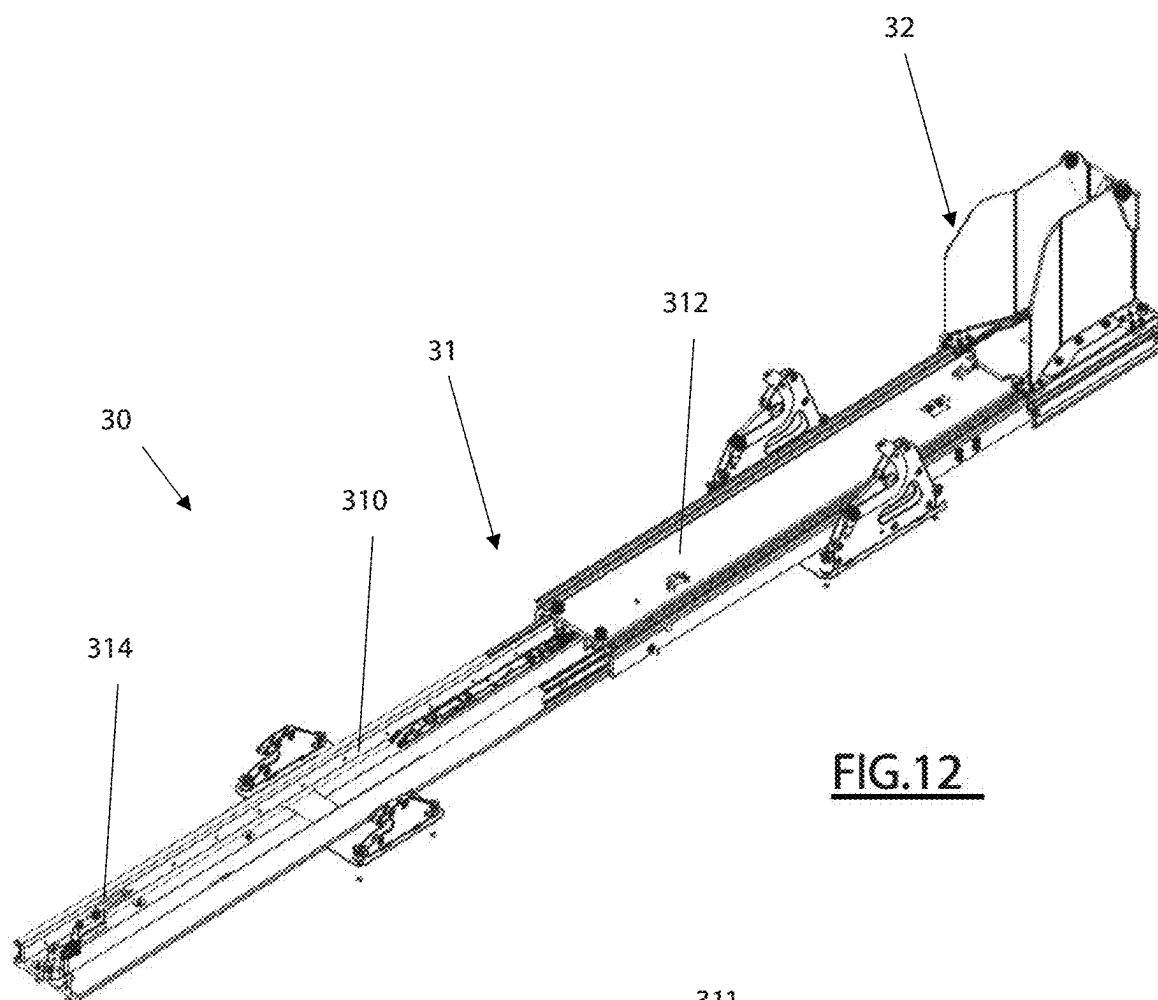


FIG. 10A







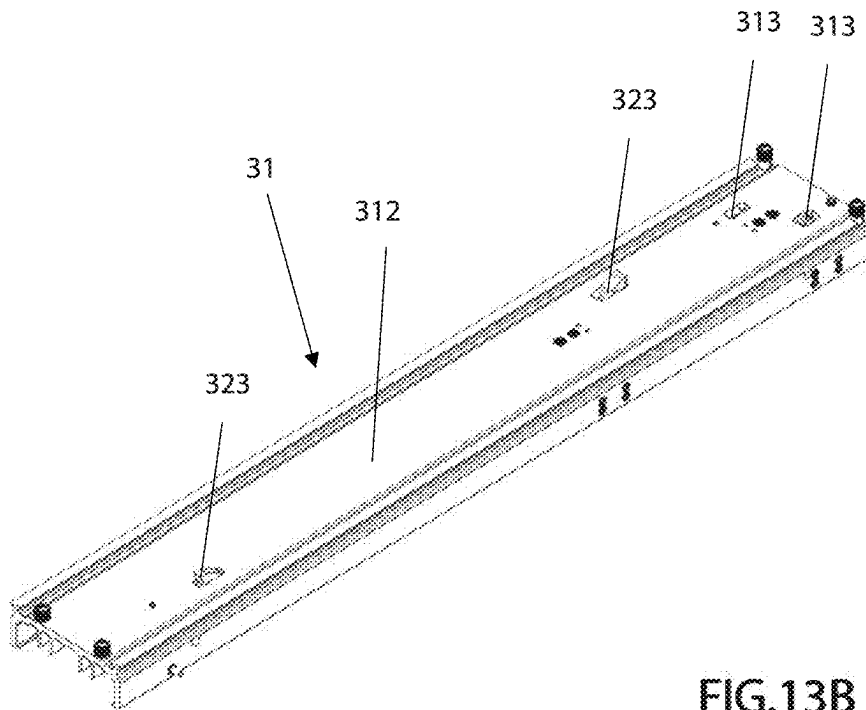


FIG.13B

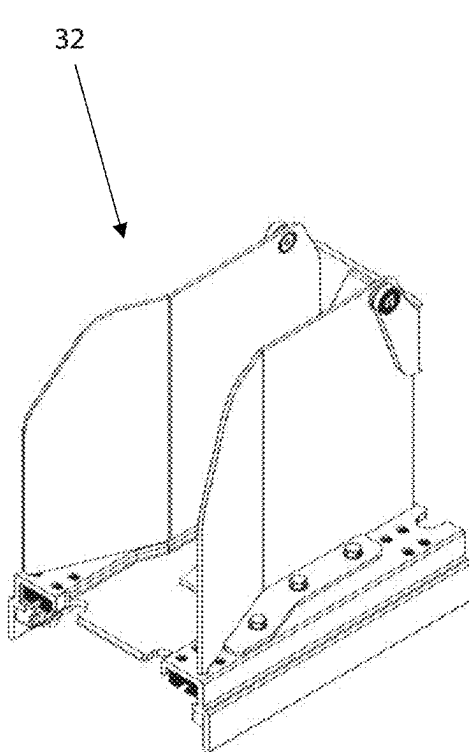


FIG.13C

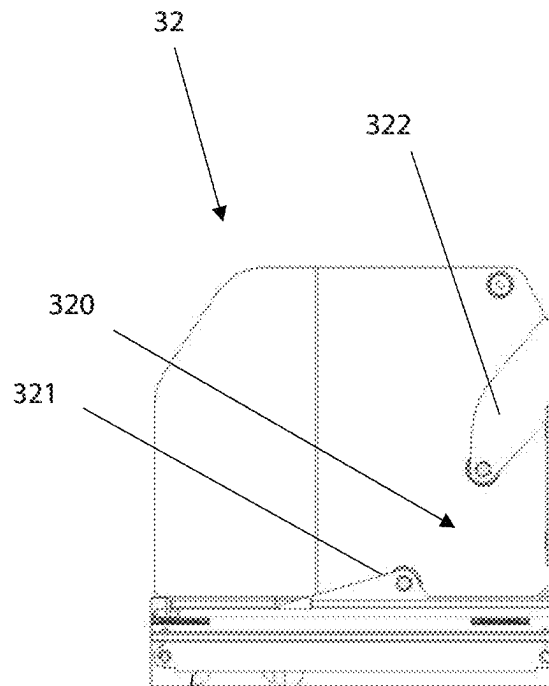


FIG.13D

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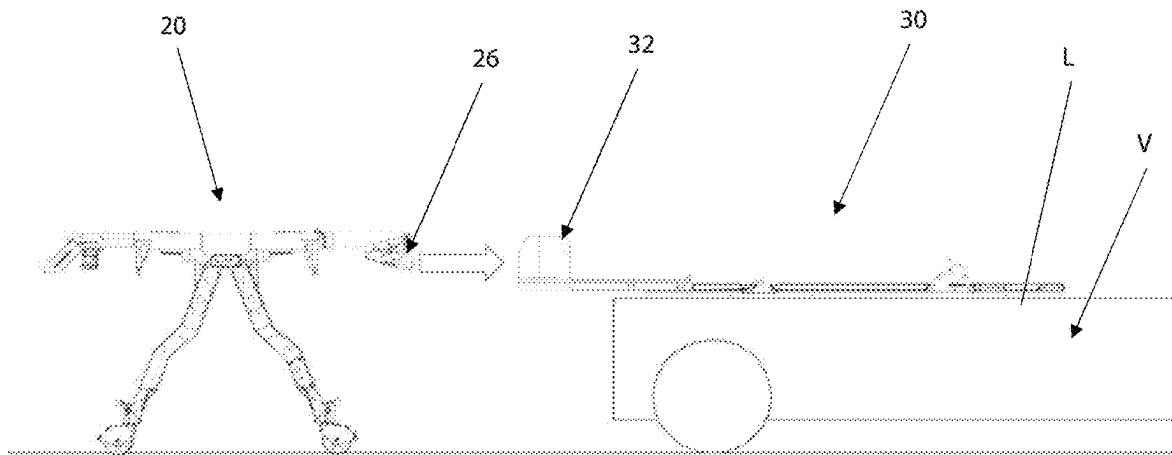


FIG. 14A

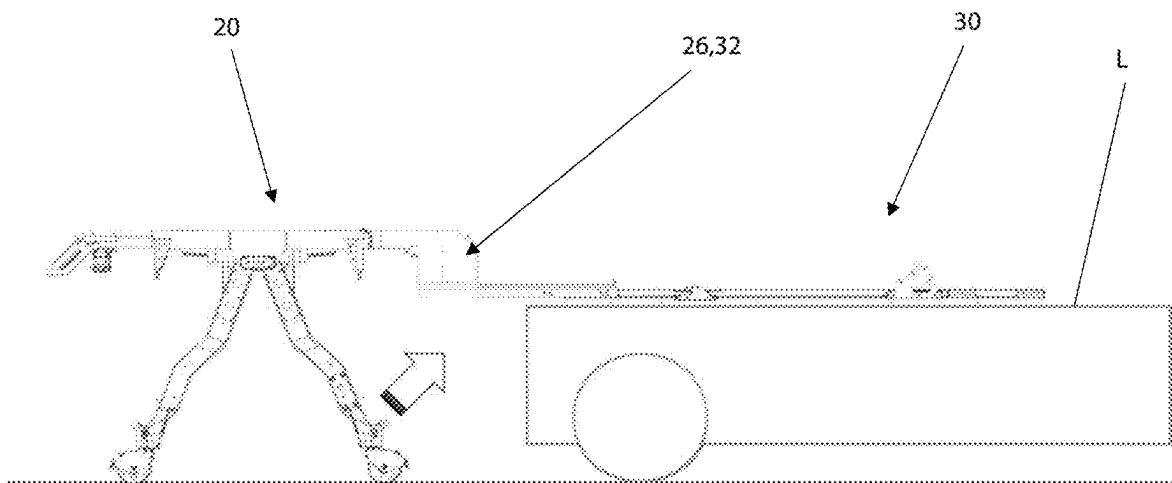


FIG. 14B

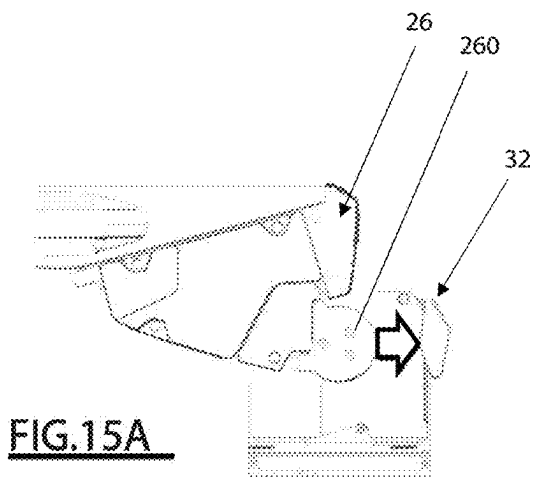


FIG. 15A

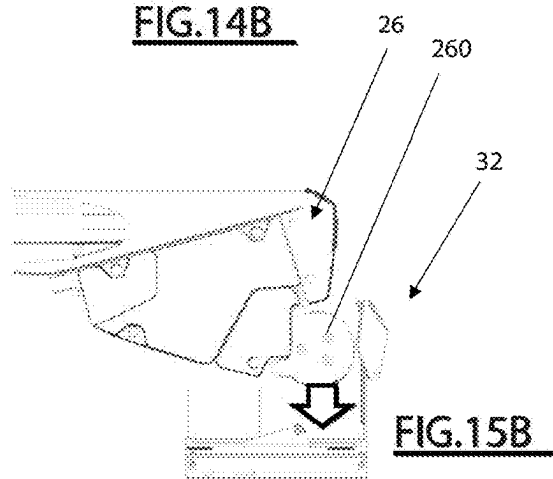


FIG. 15B

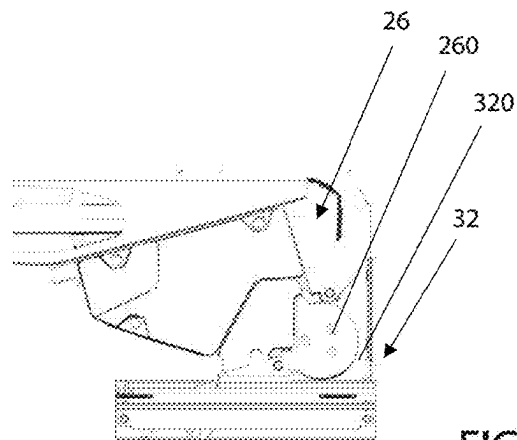


FIG. 15C

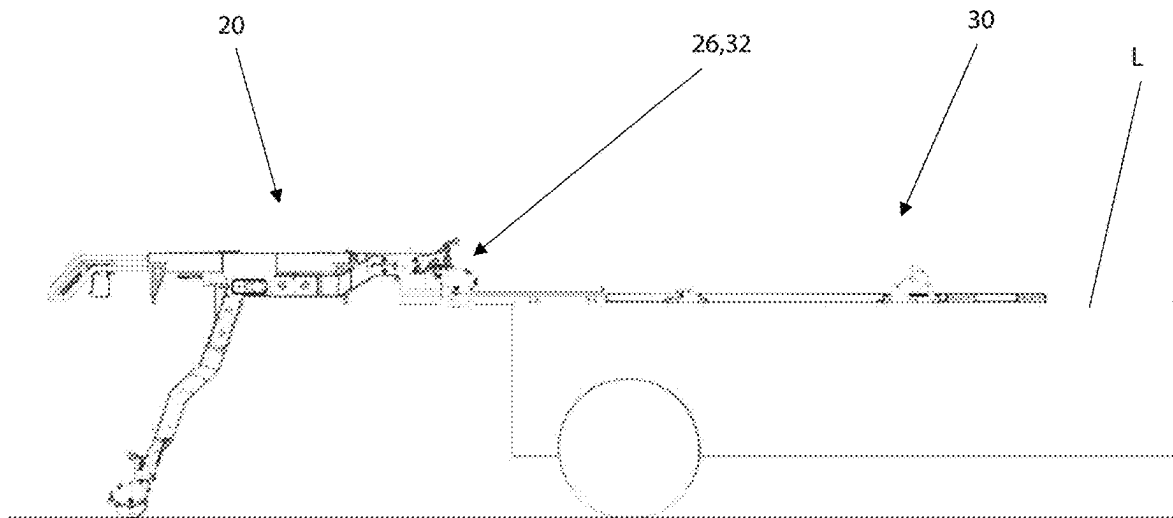


FIG. 14C

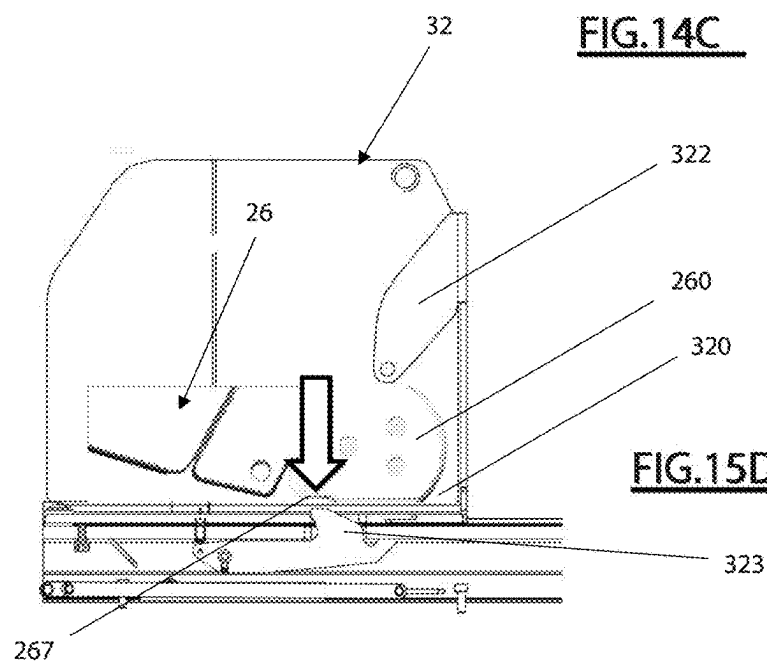


FIG. 15D

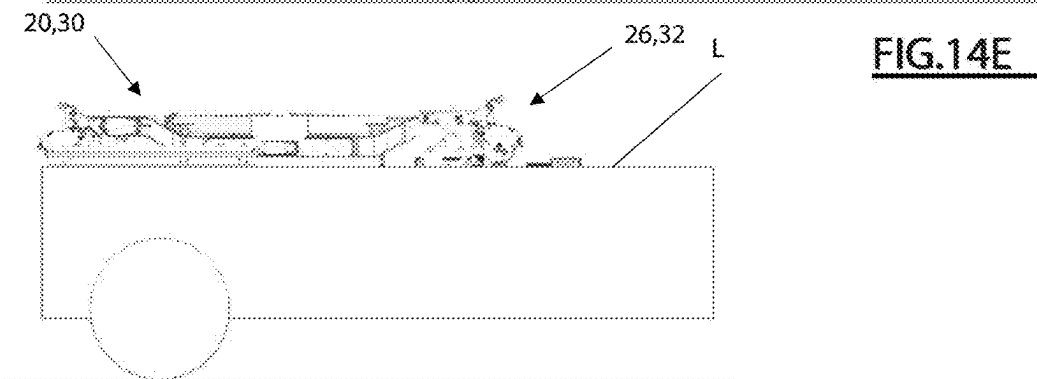
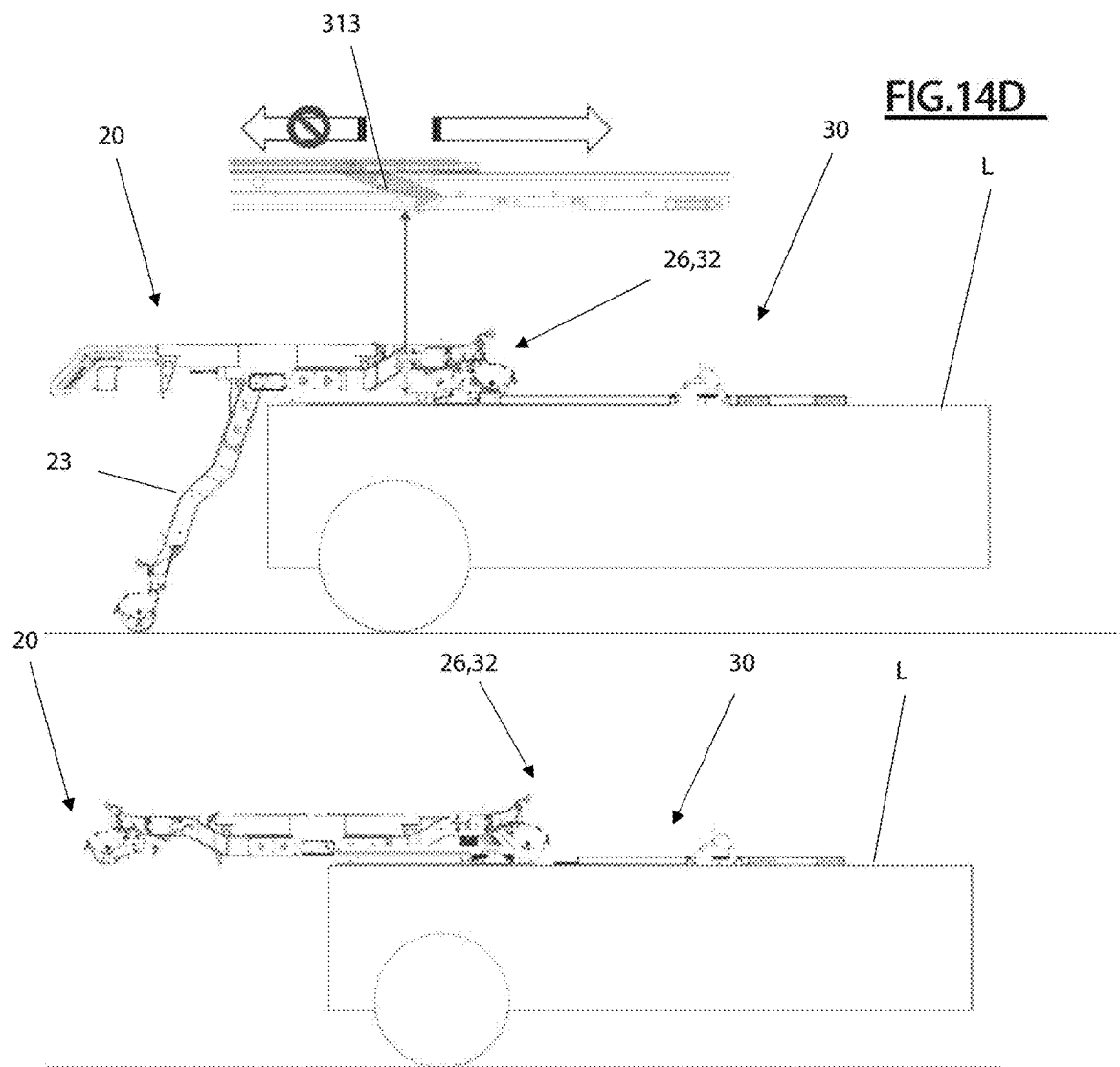


FIG. 14F

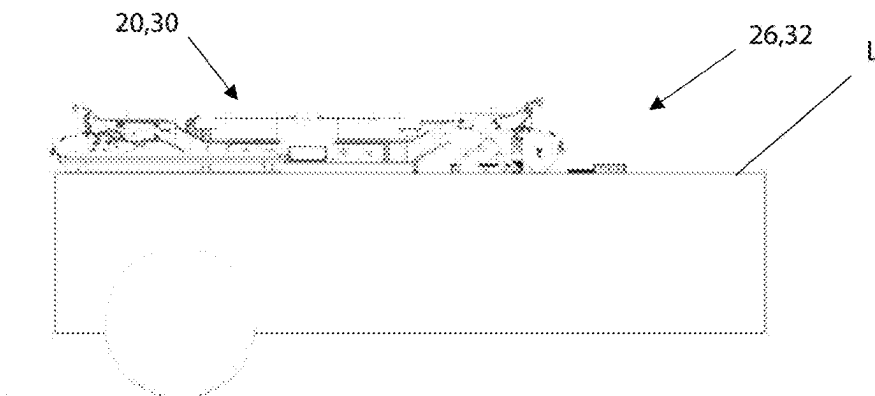


FIG. 16A

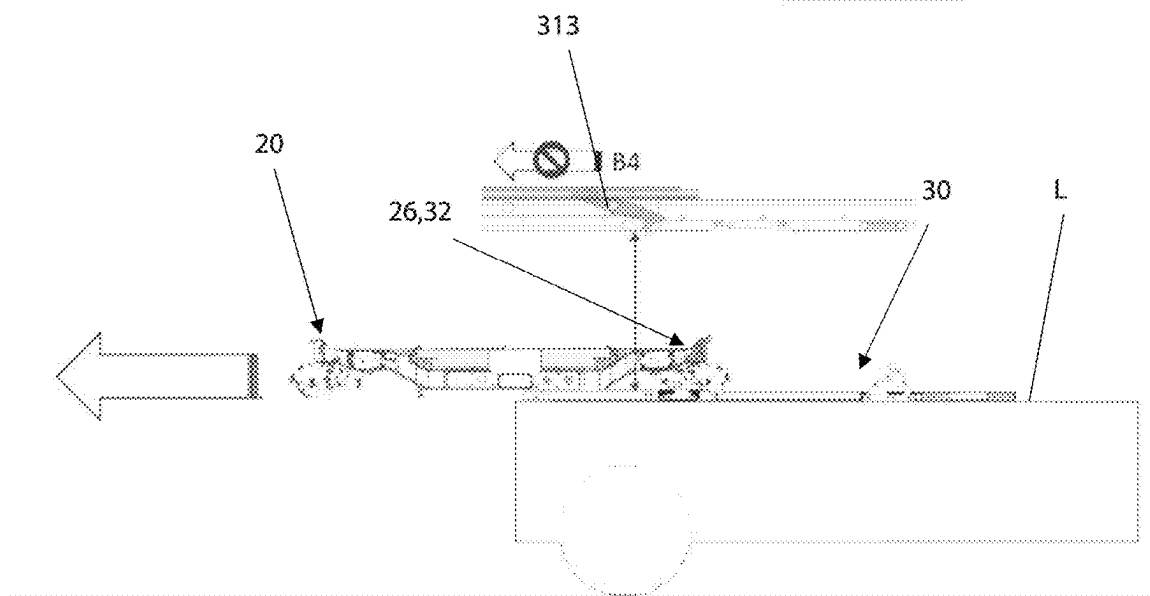


FIG. 16B

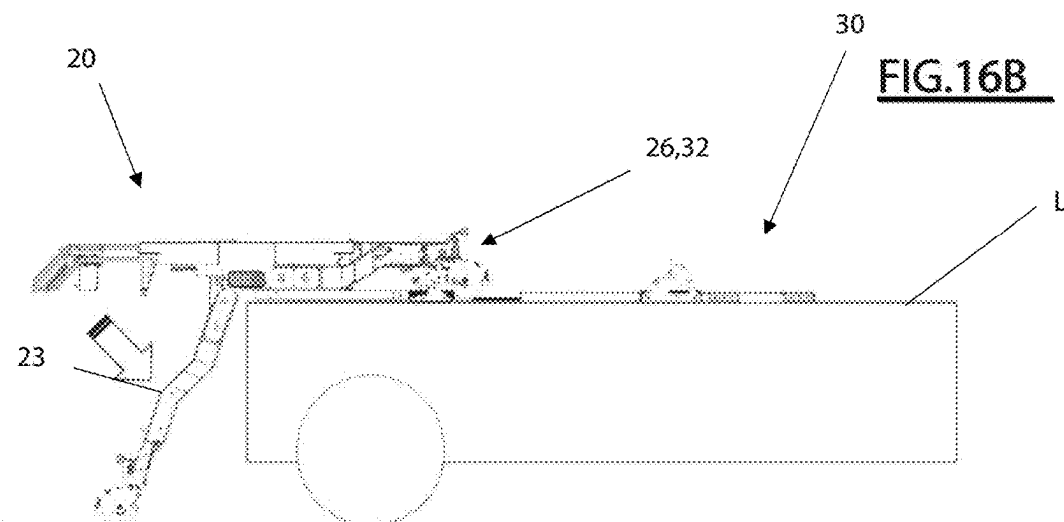
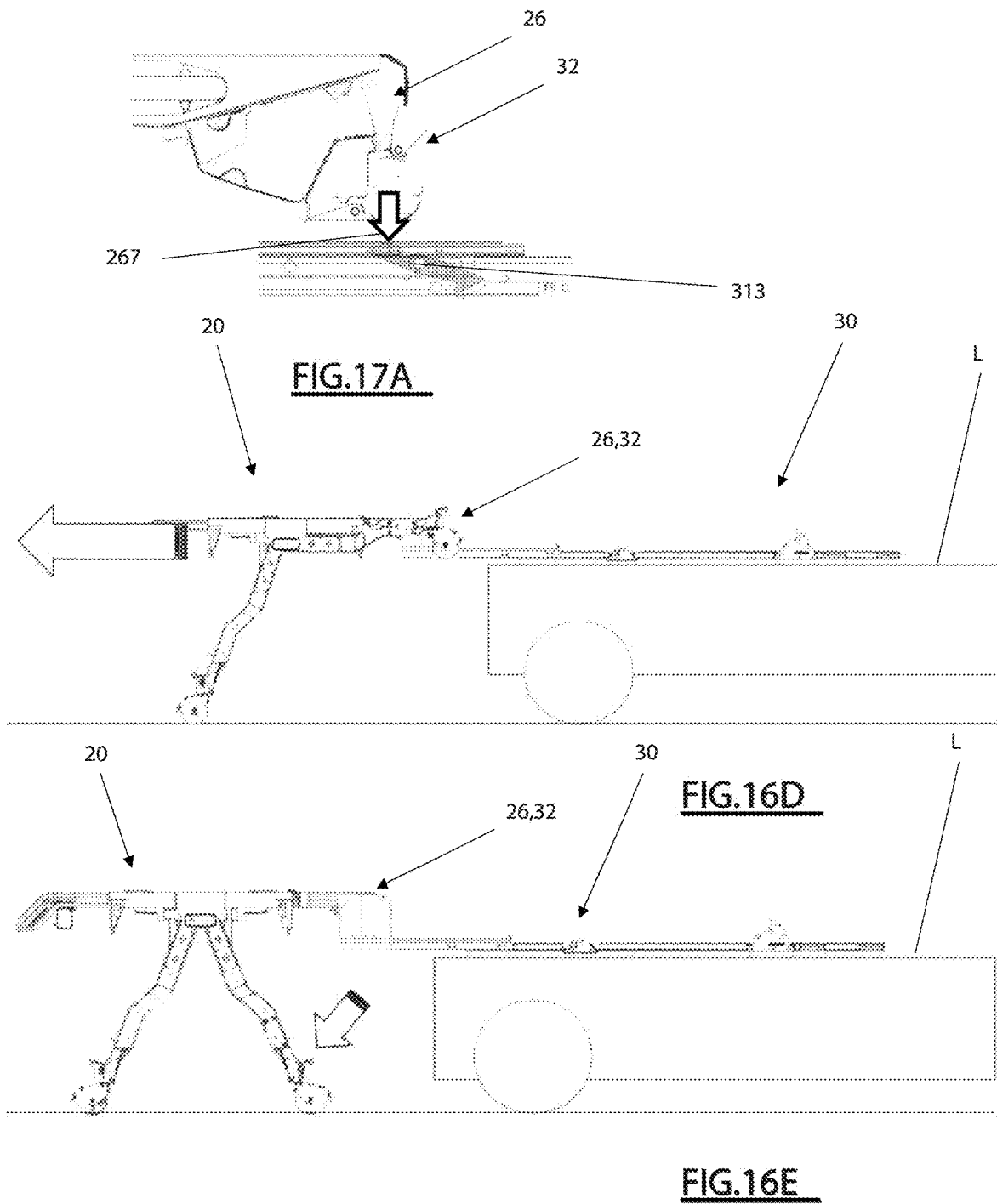


FIG. 16C



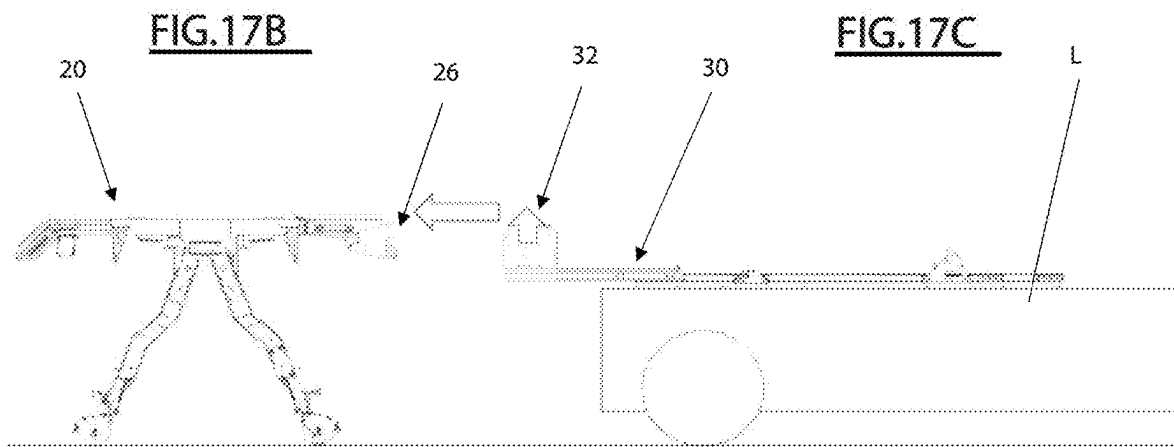
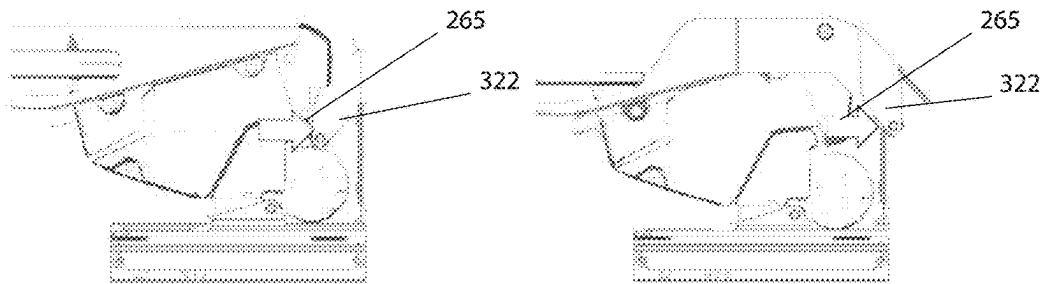


FIG. 16F

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STRETCHER

TECHNICAL FIELD

The present invention relates to a stretcher (for transporting and/or loading/unloading patients), preferably an ambulance stretcher and a relative system for loading/unloading the stretcher onto/from an ambulance loading plane.

More particularly, the present invention a stretcher of an automatic, semi-automatic or power-assisted type and the relative automatic, semi-automatic or power-assisted loading/unloading system.

PRIOR ART

As is well known, there is a variety of stretchers for transporting patients in emergencies in use. Such stretchers can be designed to transport and load/unload patients onto/from an ambulance.

Known stretchers have a fair degree of automation that allows to facilitate the loading/unloading operations of the patient onto/from the ambulance for the operator in charge and to control these operations in order to make them as safe as possible.

A need felt in the industry is to improve the safety of such loading/unloading operations, as well as to facilitate and alleviate the tasks of the loading/unloading personnel, e.g. by allowing such loading/unloading and transport operations to be carried out by a single operator and/or by relieving the operator of the burden of supporting the stretcher and the relative loads during the entire loading/unloading operation.

In addition, a further need felt in the sector is to facilitate and improve the functionality of the stretcher during the phases of transporting the patient through this stretcher, for example by making it suitable, safe and convenient for use in multiple transport conditions, for example also near ground slopes or other working situations.

An object of the present invention is to satisfy these and other needs of the prior art, within the framework of a simple, rational and low cost solution.

These objects are achieved by the features of the invention set forth in the independent claim. The dependent claims outline preferred and/or particularly advantageous aspects of the invention.

DISCLOSURE OF THE INVENTION

In order to satisfy one or more of the said needs of the prior art, the invention, in particular, makes available a stretcher, which comprises:

- a support frame for supporting a patient;
- a pair of front legs rotatably coupled to the support frame around a first rotation axis, wherein each front leg supports a front wheel holder frame hinged to the respective front leg around a first oscillation axis parallel to the first rotation axis and supporting a front wheel resting on a rest plane, wherein the front wheel pivots around a respective first pivot axis orthogonal to the first oscillation axis;
- a pair of rear legs rotatably coupled to the support frame around a second rotation axis, wherein each rear leg supports a rear wheel holder frame hinged to the respective rear leg around a second oscillation axis parallel to the second rotation axis and supporting a rear wheel resting on a rest plane, wherein the front wheel pivots around a respective second pivot axis orthogonal to the second oscillation axis;

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a handling arrangement for the front wheel holder frame and the rear wheel holder frame configured to vary, preferably independently, the inclination of the front wheel holder frame around the first oscillation axis and of the rear wheel holder frame around the second oscillation axis;

a (electronic) control module operatively connected to the handling arrangement and configured to actuate the handling arrangement in such a way as to keep the first pivot axis and the second pivot axis always orthogonal to the rest plane (i.e. to the plane on which the front wheels and the rear wheels are intended to rest and roll), for example for any angular position is assumed by one between the pair of front legs, around the first rotation axis, and the pair of rear legs, around the second rotation axis, with respect to the other between the pair of rear legs, around the second rotation axis, and the pair of front legs, around the first rotation axis.

Advantageously, the control module may comprise or be associated with a sensor arrangement, including

at least one first front angle sensor associated with the pair of front legs, wherein the first front angle sensor is configured to detect an angular position of the pair of front legs with respect to the support frame;

at least one first rear angle sensor associated with the pair of rear legs, wherein the first rear angle sensor is configured to detect an angular position of the pair of rear legs with respect to the support frame;

at least one second front angle sensor associated with at least one front wheel holder frame, wherein the second front angle sensor is configured to detect an angular position of the respective front wheel holder frame with respect to the respective front leg; and

at least one second rear angle sensor associated with at least one rear wheel holder frame, wherein the second rear angle sensor is configured to detect an angular position of the respective rear wheel holder frame with respect to the respective rear leg.

Again, the sensor arrangement may also comprise:

at least one front absolute potentiometer, associated with at least one between the pair of front legs and the front wheel holder frame, and at least one rear absolute potentiometer, associated with at least one between the pair of rear legs and a rear wheel holder frame.

Advantageously, the stretcher may comprise an actuation arrangement provided with a front first actuator, which moves the pair of front legs and which interconnects the support frame and the pair of front legs, and a rear first actuator, which moves the pair of rear legs and which connects the support frame and the pair of rear legs.

Again, the handling arrangement may comprise:

a second front actuator for each front leg of the pair of front legs, wherein each second front actuator moves the front wheel holder frame and interconnects the respective front leg of the pair of front legs and the respective front wheel holder frame; and

a second rear actuator for each rear leg of the pair of rear legs, wherein each second rear actuator moves the rear wheel holder frame and interconnects the respective rear leg of the pair of rear legs and the respective rear wheel holder frame.

Advantageously, the control module may comprise an electronic control unit operatively connected to the sensor arrangement, the second front actuator and the second rear actuator, wherein the electronic control unit is configured to perform the steps of:

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detecting a first front angle value by means of the first front angle sensor and a first rear angle value by means of the first rear angle sensor;

determining a first front compensation angle value and a first rear compensation angle value on the basis of the first front angle value and of the first rear angle value detected; and

commanding each second front actuator to perform a compensation rotation of the respective front wheel holder frame by the determined first front compensation angle value and/or each second rear actuator to perform a compensation rotation of the respective rear wheel holder frame by the determined first rear compensation angle value.

Preferably, the determination step may comprise calculating the first front compensation angle value and the second rear compensation angle value by means of the following formula:

$$\gamma = 90^\circ - \beta - \xi; \text{ and}$$

$$\theta = 90^\circ - \alpha + \xi$$

wherein γ is the first rear compensation angle, β is the first rear angle value, θ is the first front compensation angle, α is the first front angle value and ξ is the inclination angle of the support frame with respect to the rest plane.

Again, the inclination angle ξ of the support frame with respect to the rest plane can be calculated as a function of the first front angle value α and of the first rear angle value β , preferably calculated with the following formula:

$$\xi = \tan^{-1}[(s \cdot \cos(\beta) + s \cdot \cos(\alpha) + i) / (s \cdot \sin(\beta) - s \cdot \sin(\alpha))],$$

wherein s is a length of each leg of the pair of front legs **22** and each leg of the pair of rear legs and i is an interaxis between the first rotation axis and the second rotation axis.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features and advantages of the invention will be more apparent after reading the following description provided by way of a non-limiting example, with the aid of the accompanying drawings.

FIG. **1** is an axonometric view of a system according to the invention.

FIG. **2** is an axonometric view of a stretcher according to the invention.

FIG. **3** is a side view of FIG. **2** with lowered legs.

FIG. **4** is a side view of FIG. **2** with raised legs.

FIG. **5** is a schematic view of the stretcher with some sensors of the sensor arrangement thereof highlighted.

FIGS. **6A** and **6B** are views in partial medial section of the stretcher in FIG. **2**.

FIGS. **7A** and **7B** are axonometric views of a portion of the stretcher legs according to the invention, in the lowered position and in the raised position respectively.

FIGS. **8A** and **8B** are sectional views of a leg (of FIG. **7A**).

FIG. **9** is an antero-inferior axonometric view of a coupling body of the stretcher according to the invention.

FIGS. **10A** and **10B** are sectional views of the coupling body in FIG. **9**.

FIGS. **11A-11D** are sectional views of the coupling body in FIG. **9** in the respective operating configurations.

FIG. **12** is an axonometric view of a loading/unloading apparatus according to the invention.

FIGS. **13A-13D** are views of details of the loading/unloading apparatus of FIG. **12**.

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FIGS. **14A-14F** are schematic views of a sequence of loading a stretcher onto the loading/unloading apparatus.

FIGS. **15A-15D** are schematic views of detail of the coupling sequence between the coupling body and the support coupling during the loading sequence.

FIGS. **16A-16F** are schematic views of a sequence of unloading a stretcher on the loading/unloading apparatus.

FIGS. **17A-17C** are schematic views of detail of the release sequence between the coupling body and the support coupling during the unloading sequence.

BEST MODE TO IMPLEMENT THE INVENTION

With particular reference to these figures, a system for loading/unloading a stretcher, indicated globally by number **20**, onto/from a loading surface **L** of an ambulance **V**, or other patient transport and emergency vehicle, has been indicated globally by **10**.

The loading area **L** of ambulance **V** is, for example, defined by the back wall of a rear loading compartment of the ambulance **V**, which is accessible at the rear through a rear opening of ambulance **V** and which extends anteriorly in a longitudinal direction along the longitudinal axis of the ambulance **V** towards a driver's cab thereof.

The stretcher **20** is a semi-automatically driven or power-assisted stretcher (or gurney) for transporting a patient on it and being loaded onto and/or unloaded from the loading compartment of the ambulance, either directly or indirectly resting on loading surface **L**.

The stretcher **20** comprises a support frame **21** comprising a front end and a rear end, an upper platform and a lower surface (in use facing the ground).

The front end is to be understood herein as the "loading end", i.e., the axial end of the support frame **21** of the stretcher **20** that is first loaded onto the loading surface **L**. The rear end is, on the other hand, the axial end of the support frame **21** of the stretcher **20** that is last loaded onto the loading surface **L**, and is to be understood as the "control end" which is the end that provides the grip and/or the commands for the operator to control the handling of the stretcher **20**. In addition, the stretcher **20** is loaded with a patient, the patient's head can be oriented proximal to the front end and the patient's feet can be oriented proximal to the rear end. Therefore, the term "head" can be used interchangeably with the term "front" and the term "foot" can be used interchangeably with the term "rear". In general, the term "patient" means any living or formerly living load such as, for example, a human being, animal or other that can be transported and/or loaded onto the stretcher **20** for loading/unloading onto/from the loading surface **L**.

The front end and/or the rear end of the support frame **21** may be axially extendable or be fixed.

The upper platform of the support frame is configured to define a rest surface (directly or indirectly) for the patient.

Preferably, the upper platform may comprise coupling means to which a transport stretcher/bed (not shown) which supports the patient usually in a lying or semi-lying position can be fixed in a releasable way.

Furthermore, at least one coupling portion **210** (or safety hook), the function of which will be explained in more detail below, protrudes from the lower surface of the support frame **21**.

In detail, at least one pair of front coupling portions **210** (mutually symmetrical with respect to a longitudinal median plane orthogonal to the upper platform of the support frame **21**) and at least one pair of rear coupling portions **210**,

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axially separated from the pair of front coupling portions **210** (and mutually symmetrical with respect to a longitudinal median plane orthogonal to the upper platform of the support frame **21**) protrude from the lower surface of the support frame **21**.

Furthermore, the support frame **21** comprises at least one handle bar **211**, for example arranged at or near the rear end of the support frame **21**.

The handle bar **211** is configured to be grasped by one or two hands of an operator to operate the pushing or pulling of the stretcher **20** and to operate a transport thereof and/or to guide it.

The stretcher **20** then comprises a pair of front legs **22** and a pair of rear legs **23** coupled inferiorly to the support frame **21** and through which the support frame **21** is supported resting on a rest plane of the stretcher **20** (defined by the ground and/or the loading surface L).

The pair of front legs **22**, one including one on the right and one on the left, are mutually integral (for example, they are rigidly connected to each other).

Preferably, the pair of front legs **22** is articulated to the support frame so that their position can be varied with respect thereto.

In detail, the pair of front legs **22** is rotatably coupled to the support frame **21** (for example at a constrained end of each front leg **22**) around a (single) first rotation axis R1, with the possibility of rotating between two opposing angular end stroke positions, including

a raised angular end stroke position, in which the pair of front legs **22** (i.e. the free ends of the front legs **22** of the pair of front legs **22**) is proximal to the support frame (i.e. a front angle between the pair of front legs **22** and the support frame **21**, i.e. its loading platform, is minimum), and

a lowered angular end stroke position, in which the pair of front legs **22** (i.e. the free ends of the front legs **22** of the pair of front legs **22**) is distal from the support frame **21** (i.e. a front angle between the pair of front legs **22** and the support frame **21**, i.e. its loading platform, is maximum).

This front angle is, however, less than 90°, e.g. comprised between 0° and 70°. Each front leg **22** supports, at its free end, a respective front wheel holder frame **220**.

The front wheel holder frame **220** is, for example, hinged to (the free end of) the respective front leg **22** around a first oscillation axis O1 parallel to the first rotation axis R1.

Each front wheel holder frame **220**, in turn, supports a respective front wheel **221** for resting and rolling on the aforesaid rest plane.

Each front wheel **221** is preferably pivoting, i.e. capable of pivoting (in a free or controlled and/or lockable manner) around a respective first pivot axis P1 orthogonal to the first oscillation axis O1.

In detail, each front wheel **221** is pivotally connected (for free rotations), around a revolution axis, to a support element **222**, for example fork-like, which is in turn rotatably connected (in order to perform 360° rotations), around the first pivot axis P1, to the front wheel holder frame **220**.

The pair of front legs **22** and the pair of rear legs **23** are independent of each other, i.e. they are movable independently with respect to the support frame **21**.

The pair of rear legs **23**, one of which on the right and one on the left, are mutually integral (for example, they are rigidly connected to each other).

Preferably, the pair of rear legs **23** is articulated to the support frame so that their position can be varied with respect thereto.

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In detail, the pair of rear legs **23** is rotatably coupled to the support frame **21** (e.g. at a constrained end of each rear leg **23**) around a (single) second rotation axis R2 (proximal to the first rotation axis R1, e.g. parallel to and separate from it or at most also coinciding), with the possibility of rotating between two opposing angular end stroke positions, including

a raised angular end stroke position, in which the pair of rear legs **23** (i.e. the free ends of the rear legs **23** of the pair of rear legs **23**) is proximal to the support frame (i.e. a rear angle between the pair of rear legs **23** and the support frame **21**, i.e. its loading platform, is minimal), and a lowered angular end stroke position, in which the pair of rear legs **23** (i.e. the free ends of the rear legs **23** of the pair of rear legs **23**) is distal from the support frame **21** (i.e. a rear angle between the pair of rear legs **23** and the support frame **21**, i.e. its loading platform, is maximum).

This rear angle is, however, less than 90°, e.g. comprised between 0° and 70°. Each rear leg **23** supports, at its free end, a respective rear wheel holder frame **230**.

The rear wheel holder frame **230** is, for example, hinged to (the free end of) the respective rear leg **230** around a second oscillation axis O2 parallel to the second rotation axis R2.

Each rear wheel holder frame **230**, in turn, supports a respective rear wheel **231** for resting and rolling on the aforesaid rest plane.

Each rear wheel **231** is preferably pivoting, i.e. capable of pivoting (in a free or controlled and/or lockable manner) around a respective second pivot axis P2 orthogonal to the second oscillation axis O2.

In detail, each rear wheel **231** is pivotally connected (for free rotations), around a revolution axis, to a support element **232**, for example fork-like, which is in turn rotatably connected (in order to perform 360° rotations), around the second pivot axis P2, to the rear wheel holder frame **230**.

The pair of front legs **22** and the pair of rear legs **23** are mutually opposed.

In particular, the front angles and the rear angles are opposed.

In other words, the free ends of the front legs **22** of the pair of front legs **22** and the free ends of the rear legs **23** of the pair of rear legs **23** are proximal to each other when the pair of front legs **22** and the pair of rear legs **23** are in the lowered angular end stroke position and the free ends of the front legs **22** of the pair of front legs **22** and the free ends of the rear legs **23** of the pair of rear legs **23** are distal to each other when the pair of front legs **22** and the pair of rear legs **23** are in the raised angular end stroke position.

For example, the free ends of the front legs **22** of the pair of front legs **22** and the free ends of the rear legs **23** of the pair of rear legs **23** are arranged proximal and/or at, respectively, the front end and the rear end of the support frame **21**, when the pair of front legs **22** and the pair of rear legs **23** are in the raised angular end stroke position.

The first rotation axis R1 and the second rotation axis R2 are close to each other (coinciding at most) and proximal to a median plane orthogonal to the (loading platform of the) support frame **21** parallel to them.

Still, the support frame **21** and/or the pair of front legs **22** and/or the pair of rear legs **23** may also provide one or more auxiliary rest wheels projecting below from the lower surface of the support frame and having a rotation axis parallel to the first rotation axis R1 and to the second rotation axis R2 and a rest directrix arranged at the same height as the rest

directrix of the front wheels **221** and of the rear wheels **231**, when they are in the raised angular end stroke position.

The stretcher **20** comprises an actuation arrangement configured to independently actuate the handling of the pair of front legs **22** and of the pair of rear legs **23**, for example between the respective raised end stroke position and the respective lowered end stroke position.

The actuation arrangement comprises a first front actuator **241**, which moves the pair of front legs **22** and which interconnects the support frame **21** and the pair of front legs **22**.

The first front actuator **241** is, for example, a linear actuator, e.g. of the hydraulic type driven by an electric motor.

The first front actuator **241** has, for example, a cylinder, one end of which is hinged to the support frame **21**, e.g. to an ear resulting from or arranged at the lower surface thereof, and a stem, one end of which is hinged to the pair of front legs **22**, e.g. to a crossbar joining them.

The hinge axes of the stem and of the cylinder are parallel (and eccentric) to the first rotation axis **R1**.

The actuation arrangement further comprises a first rear actuator **242**, which moves the pair of rear legs **23** and which interconnects the support frame **21** and the pair of rear legs **23**.

The first rear actuator **242** is, for example, a linear actuator, e.g. of the hydraulic type driven by an electric motor.

The first rear actuator **242** has, for example, a cylinder, one end of which is hinged to the support frame **21**, e.g. to an ear resulting from or arranged at the lower surface thereof, and a stem, one end of which is hinged to the pair of rear legs **23**, e.g. to a crossbar joining them.

The hinge axes of the stem and cylinder are parallel (and eccentric) to the second rotation axis **R2**.

The stretcher **20** also comprises a handling arrangement configured to independently actuate the handling of each of the front wheel holder frame **220** around the first oscillation axis **O1** and of each rear wheel holder frame **230** around the second oscillation axis **O2** (to vary the inclination with respect to the respective leg).

The handling arrangement comprises, for each front leg **22** of the pair of front legs **22** a respective second front actuator **251**.

Each second front actuator **251** moves a respective front wheel holder frame **220** and interconnects the respective front leg **22** of the pair of front legs **22** and the respective front wheel holder frame **220**.

Each second front actuator **251** is for example a linear actuator, for example of the electric type provided with an electrically controlled brake.

Each second front actuator **251** has, for example, a cylinder, one end of which is fixed or hinged to the respective front leg **22** (e.g., internally therein), and a stem, one end of which is hinged to the respective front wheel holder frame **220**, for example, at a connection ear thereof.

The hinge axis of the stem is parallel (and eccentric) to the first oscillation axis **O2**.

In addition, the handling arrangement comprises, for each rear leg **23** of the pair of rear legs **23**, a respective second rear actuator **252**.

Each second rear actuator **252** moves a respective rear wheel holder frame **230** and interconnects the respective rear leg **23** of the pair of rear legs **23** and the respective rear wheel holder frame **230**.

Each second rear actuator **252** is for example a linear actuator, for example of the electric type provided with an electrically controlled brake.

Each second rear actuator **252** has, for example, a cylinder, one end of which is fixed or hinged to the respective rear leg **23** (e.g., internally therein), and a stem, one end of which is hinged to the respective rear wheel holder frame **230**, for example, at a connection ear thereof.

The hinge axis of the stem is parallel (and eccentric) to the second oscillation axis **O2**.

The stretcher **20** comprises a front coupling body **26** connected to the front end of the support frame **21**, e.g., facing the front and/or the bottom thereof.

The coupling body **26** comprises a coupling head **260** (facing frontally and/or inferiorly the support frame **21**), which is for example supported by a small support frame **261** rigidly fixed to the support frame **21**.

The small support frame **261** is of the box type with the coupling head **260** protruding from the front free end.

Preferably, the coupling head **260** is defined/constituted by a spherical or hemispherical (or at most truncated conical/pyramidal) body.

The coupling body **26**, in particular the coupling head **260**, is arranged on the longitudinal median plane orthogonal to the rest platform (i.e. vertical) of the support frame **21**.

Preferably, the coupling head **260** is centred on said longitudinal median plane, i.e. it has a centre that belongs to said longitudinal median plane.

Advantageously, the coupling head **260** is associated with the small support frame **261** and, therefore, with the support frame **21** in a movable manner (free to move, not actuated).

In particular, the coupling head **260** is associated with the small support frame **261** and, therefore, with the support frame **21** with the possibility of movement with respect to at least a first degree of translational (and/or roto-translational) freedom substantially parallel to the (rest platform of the) support frame **21** and, preferably, directed along the longitudinal axis of the support frame **21**, between two horizontal (mechanical) end stroke positions, including a front end stroke, wherein the coupling head **260** is distal from the support frame **21**, and a rear end stroke, wherein the coupling head **260** is proximal to the support frame **21**.

Furthermore, the coupling head **260** is associated with the small support frame **261** and, therefore, with the support frame **21** with the possibility of movement with respect to at least a second degree of translational (and/or roto-translational) freedom substantially orthogonal to the (rest platform of the) support frame **21**, between two vertical (mechanical) end stroke positions, including a lower end stroke, wherein the coupling head **260** is distal from the support frame **21**, and an upper end stroke, wherein the coupling head **260** is proximal to the support frame **21**.

In particular, the coupling head **260** is connected to the small support frame **261** by means of an articulation, which is for example defined by an articulated kinematic mechanism **262** (such as an articulated quadrilateral), which allows the translation of the coupling head **260** with respect to the aforesaid first degree of translational freedom and to the second degree of translational freedom.

The articulated kinematic mechanism **262** is defined by a plurality of levers hinged to each other (and interconnected with the small support frame **261** defining one of said levers) by means of respective articulation axes, wherein the articulation axes of the articulated kinematic mechanism are all parallel to each other and parallel to the first rotation axis **R1** and to the second rotation axis **R2**.

Advantageously, the coupling head **260** is movable from the front end stroke to the rear end stroke in contrast to first elastic means, for example defined by a first spring **263**, for example helical.

In practice, the first spring **263** is configured so as to define the front end stroke position as a stable equilibrium position for the coupling head **260** (and the rear end stroke position as an unstable equilibrium position for the coupling head **260**).

The first spring **263** is connected to the articulated kinematic mechanism, e.g. interconnected between two levers thereof.

In addition, the coupling head **260** is movable from the lower end stroke to the upper end stroke in contrast to second elastic means, e.g. defined by a second spring, e.g. helical.

In practice, the second spring is configured so as to define the lower end stroke position as a stable equilibrium position for the coupling head **260** (and the upper end stroke position as an unstable equilibrium position for the coupling head **260**).

The second spring is connected to the articulated kinematic mechanism, e.g. interconnected between two levers thereof.

Preferably, the second spring coincides with the first spring **263**.

The stretcher **20**, i.e., the coupling body **26**, further comprises a release arrangement arranged at the front end of the support frame **21**, i.e., the small support frame **261**, and configured to operate a release of the coupling body **26**, as further described below.

The release arrangement comprises, for example, a first pin **265**, slidably associated with the small support frame **261** and, therefore, with the support frame **21** along a sliding direction parallel to the (rest platform of the) support frame **21** and directed along the longitudinal axis of the support frame **21**, between two horizontal end stroke positions, including an extracted position, wherein the first pin **265** protrudes at least partially externally to the small support frame **261**, preferably beyond at least an axial portion of the coupling head **260** (at least when this is in the rear end stroke position), and distal from the support frame **21**, and a retracted position, wherein, for example, the first pin **265** retracts internally to the small support frame **261** (receding with respect to the coupling head **260**).

For example, the first pin **265** is actuated between its extracted position and its retracted position by a first actuator means, defined for example by a first servomotor **266** fixed to the small support frame **261**, for example internally thereto.

The release arrangement comprises, for example, a second pin **267**, slidably associated with the small support frame **261** and, therefore, with the support frame **21** along a sliding direction orthogonal to the (rest platform of the) support frame **21**, between two vertical end stroke positions, including an extracted position, wherein the second pin **267** at least partially protrudes externally to the small support frame **261** (inferiorly thereto), preferably beyond at least a radial portion of the coupling head **260** (at least when this is in the upper end stroke position), and distal from the support frame **21**, and a retracted position, wherein for example the second pin **267** retracts internally to the small support frame **261** (receding with respect to the coupling head **260**).

For example, the second pin **267** is actuated between its extracted position and its retracted position by a second

actuator means, defined for example by a second servomotor **268** fixed to the small support frame **261**, for example internally thereto.

The stretcher **20** comprises a sensor arrangement (stretcher sensors).

The sensor arrangement, for example, comprises at least one first front angle sensor **S1** associated with the pair of front legs **22** (and/or with the first front actuator **241**), wherein the first front angle sensor is configured to detect an angular position of the pair of front legs **22** with respect to the support frame **21**.

The sensor arrangement, for example, comprises at least one first rear angle sensor **S2** associated with the pair of rear legs **23** (and/or with the first rear actuator **242**), wherein the first rear angle sensor **S2** is configured to detect an angular position of the pair of rear legs **23** with respect to the support frame **21**.

The sensor arrangement, for example, may comprise at least one second front angle sensor **S3** associated with at least one front wheel holder frame **220**, for example one for each front wheel holder frame **220**, wherein each second front angle sensor **S3**, is configured to detect an angular position of the respective front wheel holder frame **220** with respect to the respective front leg **22**.

The sensor arrangement, for example, may comprise at least one second rear angle sensor **S4** associated with at least one rear wheel holder frame **230**, for example one for each rear wheel holder frame **230**, wherein the second rear angle sensor **S4** is configured to detect an angular position of the respective rear wheel holder frame **230** with respect to the respective rear leg **23**.

The sensor arrangement, for example, comprises at least one front absolute linear potentiometer **S5** associated with at least one front wheel holder frame **220**, for example one for each front wheel holder frame **220**, wherein each front absolute linear potentiometer **S5** is configured to detect an absolute angular position of the respective front wheel holder frame **220**.

The sensor arrangement, for example, comprises at least one rear absolute linear potentiometer **S6** associated with at least one rear wheel holder frame **230**, for example one for each rear wheel holder frame **230**, wherein each rear absolute linear potentiometer **S6** is configured to detect an absolute angular position of the respective rear wheel holder frame **230**.

The sensor arrangement, for example, comprises a first distance sensor **S7** (e.g. of the laser, on/off type) fixed to the support frame **21**, e.g. to the lower surface thereof (preferably at the transverse median plane orthogonal to the longitudinal axis of the support frame), facing downwards, wherein the first distance sensor **S7** is configured to detect a distance between the support frame **21** (i.e. its lowest lower surface) and the underlying rest plane.

The sensor arrangement, for example, comprises a second distance sensor **S8** (e.g. of the laser, on/off type) fixed to the support frame, e.g., near the front end thereof, preferably at the small support frame **261**, e.g. at the lower surface thereof, facing downwards, wherein the second distance sensor **S8** is configured to detect a distance between the support frame **21**, i.e. the small support frame **261** (i.e. its lowest lower surface) and an underlying abutment surface.

The sensor arrangement, for example, comprises a first proximity sensor **S9** (e.g. of the magnetic type) fixed to the support frame, for example, near the front end thereof, preferably at the front-facing small support frame **261**, wherein the first proximity sensor **S9** is configured to detect

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a proximity between the support frame **21**, i.e. the small support frame **261**, and a front abutment surface.

The sensor arrangement, for example, comprises a first limit switch sensor **S10**, for example fixed to the small support frame **261**, which is configured to detect when the coupling body **26**, i.e. the coupling head **260**, is in its rear end stroke position.

For example, the first limit switch sensor **S10** is of the type of a contact (mechanical) switch (of the on/off type).

The sensor arrangement, for example, comprises a second limit switch sensor **S11**, for example fixed to the small support frame **261**, which is configured to detect when the coupling body **26**, i.e. the coupling head **260**, is in its upper end stroke position.

For example, the second limit switch sensor **S11** is of the type of a contact (mechanical) switch (of the on/off type).

The sensor arrangement, for example, comprises a third distance sensor **S12** (e.g. of laser type), arranged on the coupling body **26**, for example integral with at least one between the small support frame **261** and the coupling head **260**.

The sensor arrangement, for example, comprises a second proximity sensor **S13** (e.g., a magnetic reed) arranged on/in proximity to at least one coupling portion **210**, for example on each front coupling portion **210** or preferably only at the rear coupling portions **210**.

The sensor arrangement, for example, comprises a front pressure sensor **S14** associated with the (hydraulic circuit of) first front actuator **241**, which is for example configured to detect a pressure value of the actuating fluid of the first front actuator **241**.

The sensor arrangement, for example, comprises a rear pressure sensor **S15** associated with the (hydraulic circuit of) first rear actuator **242**, which is for example configured to detect a pressure value of the actuating fluid of the first rear actuator **242**.

The sensor arrangement, for example, comprises an inclinometer **S16** associated with the support frame **21**, e.g. at/in proximity to the rear end thereof, as further described below.

The stretcher **20** further comprises a power supply system on board the stretcher.

For example, the stretcher **20** comprises at least one battery (or battery pack) fixed to the support frame **21**, for example in a rechargeable and/or removable and/or replaceable manner.

The supply system is configured to supply power to the actuation arrangement, and/or the handling arrangement and/or the sensor arrangement and/or the release arrangement and/or a control module (described hereinbelow).

The stretcher **20** further comprises a (electronic) control module **27**, which is, for example, arranged at/in proximity to the rear end of the support frame **21**.

The control module **27** is, generally, configured to receive commands as input from the operator and provide indications as output to be made available to the operator and/or other command signals to be made available to the system **10** and/or to the stretcher **20**.

The control module **27**, for example, may comprise one or more commands **270** which can be actuated by the operator.

For example, the commands **270** can be fixed to the handle bar **211** and/or near it at the rear end of the support frame **21**.

The operator can use the commands **270** in the loading and unloading of the stretcher **20** to control and/or command the movement of the pair of front legs **22** and of the pair of rear legs **23** and other.

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The commands **270** may further comprise one or more lifting buttons (“+”) which can be actuated to raise the stretcher **20** and one or more lowering buttons (“-”) which can be actuated to lower the stretcher **20**.

Each of the lifting buttons and the lowering buttons may generate signals that actuate the pair of front legs **22**, the pair of rear legs **23** or both to perform functions of the stretcher **20**, which provide for the pair of front legs **22**, the pair of rear legs **23** or both to be lowered or raised.

In some embodiments, each of the lifting buttons and of the lowering buttons may be analogue (i.e., pressing and/or moving the button may be proportional to a parameter of the control signal).

The actuation speed of the pair of front legs **22**, of the pair of rear legs **23** or both can be proportional to the control signal parameter.

The control module **27** may comprise a visual display component or graphical user interface **271** configured to make (visual, tactile, auditory or other) information available to the operator.

For example, the user interface **271** is fixed to the rear end of the support frame **21**.

The user interface **271** may comprise any device capable of emitting an image such as, for example, a liquid crystal display, a touch screen or the like.

One or more lifting buttons and lowering buttons can be defined as integral to the graphical interface.

In addition, the inclinometer **S16** can be defined as integrated in the graphical interface.

The stretcher **20**, i.e. its control module **27**, further comprises an electronic control unit **272** (of the stretcher).

The electronic control unit **272** may be any device/processor capable of executing machine-readable instructions such as, for example, a controller, an integrated circuit, a microchip or the like.

As used herein, the term “communicatively coupled” means that the components are capable of exchanging data signals with each other such as, for example, electrical signals via conductive medium, electromagnetic signals via air, optical signals via optical waveguides and the like.

The electronic control unit **272** may be provided with or connected to one or more memory modules, which may be any device capable of storing data and/or instructions and/or software programmes that can be read and implemented by the electronic control unit **272**.

The electronic control unit **272** is operatively connected to the actuation arrangement, and/or the handling arrangement and/or the sensor arrangement and/or the control module **27** and/or the supply system and/or the release arrangement.

The system **10** further comprises a loading/unloading apparatus **30**, which is fixed to or carried by the ambulance **V**.

The loading/unloading apparatus comprises a longitudinal guide **31**, which is configured to be placed on the loading surface **L** of the ambulance **V** (parallel to the longitudinal axis of the ambulance).

The guide **31** comprises, for example, a fixed rail **310**, which is fixed (e.g. bolted) to the loading surface **L**.

The fixed rail **310** has a length substantially equal to the axial length of the stretcher **20**.

Further, the fixed rail **310** has a rear end arranged at or proximal to the rear opening of the ambulance **V** and an opposing front end arranged proximal to the driver’s cab of the ambulance **V**.

Safety couplings **311** (so-called hooks **20g**) rise from the fixed rail **310** and are configured to couple to the coupling portion **210** of the stretcher **20**.

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In detail, at least one pair of front safety couplings **311** protrude from the upper surface of the fixed rail **310**, i.e. they are distal from the rear opening of the ambulance **V**, which are (mutually symmetrical with respect to a median longitudinal plane orthogonal to the loading plane **L** and) configured to couple (snap-fittingly) to the pair of front coupling portions **210** of the stretcher **20**, and at least one pair of rear safety couplings **311**, i.e. proximal to the rear opening of the ambulance **V**, which are (mutually symmetrical with respect to the median longitudinal plane orthogonal to the loading plane **L** and) configured to couple (snap-fittingly) to the pair of rear coupling portions **210** of the stretcher **20**.

The guide **30**, for example, may comprise an intermediate slide **312**, which is slidingly (superiorly) associated with the fixed rail **310**, along a sliding direction parallel to the longitudinal axis of the fixed rail **310**.

The slide **312**, for example, is essentially half long the length of the fixed rail **310**.

The slide **312** has a rear end arranged proximal to the rear opening of the ambulance **V** and an opposing front end arranged proximal to the driver's cab of the ambulance **V**.

For example, the slide **312** is configured to slide (with free sliding, i.e. not actuated) along the fixed rail **310** between two end positions, including a front end position, wherein for example the front end of the slide **312** is substantially placed at the front end of the fixed rail **310**, and a rear end position, wherein for example the rear end of the slide **312** projects axially with respect to the rear end of the fixed rail **310** (by a stretch substantially equal to half the length of the slide **312**), preferably so as to be able to project substantially outside the loading surface **L** (and therefore the loading compartment) of the ambulance **V**.

Between the two end positions, the slide **312** travels substantially $\frac{3}{4}$ of the length of the fixed rail **310**.

The guide **31** further comprises one or more coupling elements **313**, interposed between the slide **312** and the fixed rail **310**, configured to (temporarily) stop the slide **312** at corresponding axial stop positions along the travel between the two end positions and/or at each of said end positions.

In particular, the guide **31** has a front coupling element **313** configured to (temporarily) stop the slide **312** in the rear end position.

Further, the guide **31** has an intermediate coupling element **313** configured to (temporarily) stop the slide **312** at an intermediate stop position between the front end position and the rear end position, for example wherein the rear end of the slide **312** is placed substantially at the rear end of the fixed rail **310**.

For example, the intermediate coupling element **313** defines a unidirectional constraint that does not allow the slide **312** to slide in the direction of approach to the rear end position (but it allows the slide **312** to slide in the direction of approach to the front end position).

For example, the coupling elements **313** are configured to be releasable from the release arrangement of the stretcher **20**, i.e., from the second pin **267** (in the passage from the retracted position to the extracted position thereof, when the second pin **267** is at, or superimposed in plan, on a release appendage of the coupling element **313** which emerges above the slide **312**, at least when it couples with the fixed rail **310**) and/or from a cam system which can be actuated by a support coupling **32** (described in detail below).

The loading/unloading apparatus **30** further comprises a support coupling **32**, which is slidingly (superiorly) connected to the guide **31** along a sliding direction parallel to the longitudinal axis of the guide.

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The support coupling **32** is configured to receive through releasable coupling the coupling body **26**, i.e., the coupling head **260**, of the stretcher **20**, as will be more fully described below, and/or to support at least partially the stretcher **20** (performing an anti-tip function for it).

The support coupling **32** defines a concave seat formed by a rear wall (orthogonal to the longitudinal axis of the guide **31**), two lead-in side walls, having a free rear end and a rear end which is joined to the rear wall, and a lower wall (which is joined to the side walls and to the rear wall).

In practice, the support coupling **32** is defined by a box-like body open at the top and front and closed laterally by the side walls, at the rear by the rear wall and at the bottom by the lower wall.

The lead-in side walls preferably converge towards the rear wall, so that the free front ends are at a greater distance apart than the distance between the rear ends.

The concave seat contained between the lead-in side walls, the rear wall and the lower wall delimits an internal volume within which a coupling seat **320** is contained.

The coupling seat **320** is configured to define a snap-on coupling, releasable, with the coupling head **260** of the stretcher **20**.

The coupling seat **320**, in this case, comprises a first lower coupling **321**, which is, for example, fixed with respect to the coupling seat **320**.

The free upper end of the first coupling **321** is, for example, associated with a revolution member, such as a roller (rotatably associated with the first coupling **321** with respect to a rotation axis parallel to the loading surface **L** and orthogonal to the sliding direction).

The roller is configured to roll on the coupling head **260** during the coupling and release operations.

The coupling seat **320** further comprises a second upper coupling **322**, which is movable with respect to the coupling seat **320**.

The second coupling **322** is, for example, movable from a rear position to a front position, for example in contrast to an elastic thrust force, preferably exerted by a thrust spring, for example helical.

Preferably, the second coupling **322** is associated in a tilting manner with the coupling seat **320**, for example with the rear wall thereof (and facing frontally therefrom), around a second (horizontal) tilting axis orthogonal to the sliding direction of the support coupling **32**.

The free upper end of the second coupling **322** is, for example, associated with a revolution member, such as a roller (rotatably associated with the second coupling **322** with respect to a rotation axis parallel to the second tilting axis).

The roller is configured to roll on the coupling head **260** during the coupling and release operations.

In practice, the coupling seat **320** is defined between the first coupling **321** and the second coupling **322** (i.e., between the two rollers) and is selectively configurable between two operating positions, including:

- a first open configuration, in which the second coupling **322** is in the rear position (and the distance between the rollers is such as to allow the passage of the maximum diameter zone of the coupling head **260**); and
- a second closed configuration, in which the second coupling **322** is in the front position (and the distance between the rollers is minimal and such as to prevent the passage of the maximum diameter zone of the coupling head **260**).

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The second coupling **322**, moreover, is such as to define an anti-tip constraint for the stretcher **20**, i.e. it is such as to oppose a vertical thrust directed upwards.

For example, the second coupling **322** is configured so as to be releasable from the release arrangement of the stretcher **20**, i.e., from the first pin **265** (in the passage from the retracted position to the extracted position thereof, when the first pin **265** is at, i.e., horizontally aligned and at a pre-determined axial distance, to a release appendage of the second coupling **322**, at least when it couples the coupling head **260**).

The support coupling **32**, for example, has an axial length substantially less than half of the length of the slide **312** to which it is fixed, for example equal to 1/3 of the length of the slide **312**.

The support coupling **32** has a front (open) end arranged proximal to the rear opening of the ambulance **V** and an opposing rear end, defined by the rear wall, arranged proximal to the driver's cab of the ambulance **V**.

For example, the support coupling **32** is configured to slide (with free sliding, i.e., not actuated) along the slide **312** between two end positions, including one front end position, wherein, for example, the rear end of the support coupling **32** is placed substantially at the front end of the slide **312**, and one rear end position, wherein, for example, the front end of the support coupling **32** is placed substantially at the rear end of the slide **312**.

Between the two end positions, the coupling element **32** travels substantially 3/4 of the length of the slide **312**.

At least one between the slide **312** and the support coupling **32** further comprises one or more coupling elements **323**, interposed between the slide **312** and the support coupling **32**, configured to (temporarily) stop the support coupling **32** in corresponding axial stop stations along the travel between the two end positions and/or at each of said end positions.

In particular, the slide **312** has a rear coupling element **323** configured to (temporarily) stop the support coupling **32** in the rear end position and a front coupling element **323** configured to (temporarily) stop the support coupling **32** in the front end position.

For example, the coupling elements **323** are configured to be releasable from the release arrangement of the stretcher **20**, i.e., from the second pin **267** (in the passage from the retracted position to the extracted position thereof, when the second pin **267** is at, i.e., superimposed in plan, to a release appendage of the coupling element **323** which emerges above the support coupling **32**, at least when it couples the slide **312**) and/or from cam elements **31** fixed to the fixed rail **310** and intended to come into contact with the coupling element **323** during the sliding of the slide **312** on the fixed rail **310** from the front end position to the rear end position.

The rear coupling element **323** is, for example, released by the second pin **267**.

The front coupling element **323** is, for example, released by means of such cam elements (i.e. a linear cam).

The loading/unloading apparatus **30** may comprise at least a first sensor configured to detect when the slide **312** is in its rear end position and/or in its front end position (with respect to the fixed rail **310**) and/or a second sensor configured to detect when the support coupling **32** is in its front end position and/or in its rear end position (with respect to the slide **312**).

In addition, the loading/unloading apparatus may comprise a further electronic control unit (not shown) also having an interface module, e.g. defined by a visual/acoustic beacon and/or configured to connect to the control module

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27 (i.e. to the electronic control unit **272** and/or to the user interface **271**) of the stretcher **20**, e.g. wirelessly.

In light of that described above, the operation of the stretcher **20** is the following.

During the handling of the stretcher **20**, with the front wheels **221** and the rear wheels **231** in rolling rest on a rest plane, whether defined by the ground or by the loading plane **L** or otherwise, whether horizontal or inclined with respect to the horizontal, the control module **27** is configured to actuate the handling arrangement, i.e., the second front actuator **251** and/or the second rear actuator **252**, so as to keep the first pivot axis **P1** and/or the second pivot axis **P2** always orthogonal to said rest plane, for example whatever angular position is assumed by one between the pair of front legs **22**, around the first rotation axis **R1**, and the pair of rear legs **23**, around the second rotation axis **R2**, with respect to the other between the pair of rear legs **23**, around the second rotation axis **R2**, and the pair of front legs **22**, around the first rotation axis **R1**.

Therefore, both when the rest plane is horizontal (and the support frame, i.e. its rest platform, is horizontal or inclined to the horizontal) and when the rest plane is inclined with respect to the horizontal (and the support frame, i.e. its rest platform, is horizontal or inclined with respect to the horizontal), the first pivot axis **P1** and the second pivot axis **P2** are always kept orthogonal to said rest plane by the control module **27**.

In particular, the control module **27** is configured to always keep the first pivot axis **P1** and the second pivot axis **P2** orthogonal to the plane in which the revolution axes of the front wheels **221** and of the rear wheels **231** lie.

In detail, the electronic control unit **272** is configured to perform the steps of:

detecting a first front angle value by means of the first front angle sensor **S1** and a first rear angle value by means of the first rear angle sensor **S2**;

determining a first front compensation angle value and a first rear compensation angle value on the basis of the first front angle value and of the first rear angle value detected; and

commanding each second front actuator **251** to perform a compensation rotation (around the first oscillation axis **O1**) of the respective front wheel holder frame **220** by the determined first front compensation angle value and/or each second rear actuator **252** to perform a compensation rotation (around the second oscillation axis **O2**) of the respective rear wheel holder frame **230** by the determined first rear compensation angle value.

Advantageously, the determination step comprises calculating the first front compensation angle value and the second rear compensation angle value by means of the following formula:

$$\gamma = 90^\circ - \beta - \xi; \text{ and} \quad d)$$

$$\theta = 90^\circ - \alpha + \xi \quad e)$$

wherein γ is the first rear compensation angle, β is the first rear angle value, θ is the first front compensation angle, α is the first front angle value and ξ is the inclination angle of the support frame with respect to the rest plane.

The inclination angle ξ of the support frame with respect to the rest plane is calculated as a function of the first front angle value α and of the first rear angle value β , preferably calculated with the following formula:

$$\xi = \tan^{-1}[(s \cdot \cos(\beta) + s \cdot \cos(\alpha) + i) / (s \cdot \sin(\beta) - s \cdot \sin(\alpha))], \quad f)$$

wherein s is a length of each leg of the pair of front legs **22** (e.g. calculated from the first rotation axis **R1** to the revo-

lution axis of the respective rear wheel **221**) and each leg of the pair of rear legs **23** (e.g. calculated from the second rotation axis **R2** to the revolution axis of the respective rear wheel **231**), and *i* is an interaxis between the first rotation axis **R1** and the second rotation axis **R2**.

The aforesaid functions/formulas are stored in the memory modules of the electronic control unit **272**.

In light of the above, the operation of the system **10** is as follows.

The electronic control unit **272** of the stretcher **20** is configured to perform (and/or assist in performing) a sequence of loading the stretcher **20** onto the loading surface **L** of the ambulance **V**, i.e., on the loading/unloading apparatus **30**.

While performing the loading sequence, the operator may (or must) hold down a loading button and/or initiate a loading sequence via the user interface **271**, the release of such a button safely locks any handling of the stretcher **20**.

First, an operator (or the electronic control unit of the loading/unloading apparatus) checks that the slide **312** is in its rear end position and locked therein by the rear coupling element **313** and the support coupling **32** is in its rear end position and locked therein by the rear coupling element **323**.

When the loading sequence is activated, first the support frame **21** of the stretcher **20** is brought to a predetermined loading height by actuating the first front actuator **241** and/or the second rear actuator **242**.

This height is configured so that the coupling head **260** is at a height greater than the lower wall of the support coupling **32** (but less than the maximum height of the rear wall thereof).

At this point, the operator guides the stretcher **20** so as to bring the coupling head **260** within the support coupling **32**, for example guided by the lead-in side walls thereof.

When the coupling head **260** enters the support coupling **32** it is pressed by the operator against the rear wall thereof (and/or against the second coupling **322**), and this pressure brings the coupling head **260** from its front end stroke to its rear end stroke.

When the coupling head **260** reaches its rear end stroke, the first limit switch sensor **S10** detects this position and, for example, the first proximity sensor **S9** recognises that the coupling head **260** is in abutment against the rear wall of the support coupling **32** (and not against an occasional obstacle), consequently, the electronic control unit **271** detects the correct positioning of the coupling head **260** in the support coupling **32** based on the (electrical) signal received by the first limit switch sensor **S10**.

At this point, the electronic control unit **272** gives its consent to the next steps of the loading sequence.

In particular, the electronic control unit **272** commands the first front actuator **241** and the first rear actuator **242** so as to lower the stretcher **20**, i.e. the support frame **21** thereof, vertically.

When the coupling head **260** is pressed by the lowering against the lower wall of the support coupling **32** within the coupling seat **320** thereof, such pressure brings the coupling head **260** from its lower end stroke to its upper end stroke.

When the coupling head **260** reaches its upper end stroke, the second limit switch sensor **S11** detects this position and, consequently, the electronic control unit **272** detects the correct positioning of the coupling head **260** in the support coupling **32** based on the (electrical) signal received by the second limit switch sensor **S11**.

In practice, the lifting of the coupling head **260** from the lower end stroke to the upper end stroke is indicative of (a

height of the support frame **21** and/or a load bearing on the coupling body **26** (i.e. on the coupling head) detected by means of the second limit switch sensor **S11**.

In fact, when the load bearing on the coupling head **260** is lower than a predetermined loading value, the coupling head **260** does not reach the upper end stroke, whereas when instead the load bearing on the coupling head **260** exceeds or equals the predetermined loading value, the coupling head **260** reaches the upper end stroke.

When the first limit switch sensor **S10** and the second limit switch sensor **S11** detect that both the rear end stroke and the upper end stroke of the coupling head **260** have been reached, the coupling head **260** has entered the coupling seat **320** and is retained therein between the first coupling **321** and the second coupling **322**.

The electronic control unit **272** is configured to query the third distance sensor **S12**, in order to verify the correct alignment/parallelism of the stretcher **20** (i.e. the support frame thereof) with respect to the guide **31**.

At this point, the electronic control unit **272**, when it receives the signal from the second limit switch sensor **S11**, is configured to operate the lifting of the pair of front legs **22** (up to the raised angular end stroke position), by actuating the first front actuator **241**, based on the indicative signal detected.

The electronic control unit **272**, moreover, is configured to determine an inclination of the support frame **21** with respect to the slide **312** (during the lifting of the pair of front legs **22**), for example by means of the third distance sensor **S12** and, to command the lifting/lowering of the pair of rear legs **23**, by actuating the first rear actuator **242**, to keep the support frame **21** parallel to the rest plane/ground (and/or to the loading surface **L**).

When the pair of front legs **22** is in its raised angular end stroke position, the electronic control unit **272** can first confirm/verify the reaching thereof by querying the first front angle sensor **S1**.

In addition, the electronic control unit **272** is configured to command and actuate the unlocking arrangement, for example by commanding the second pin **267** to move to its extracted position, so as to unlock the locking arrangement, i.e. the rear coupling element **232** (to allow the support coupling **32** to slide on the slide **312** towards its front end position).

The operator can then push the stretcher **20** forward.

As soon as the advancement of the stretcher **20** begins, the electronic control unit **272** can verify that the slide **312** is free to slide on the fixed rail **310**, for example by means of the second distance sensor **S8**, and commands the second pin **267** to return to its retracted position.

At this point, the operator can push the stretcher **20** horizontally until the support coupling **32** reaches its front end position.

When the support coupling **32** reaches its front end position on the slide **312** it unlocks (by means of a special mechanism) the coupling element **313** allowing the slide **312** to slide on the fixed rail **310** (from the front end position to the rear end position).

At this point, the electronic control unit **272** checks the position of the stretcher with respect to the loading surface **L**, in particular, it queries the first distance sensor **S7**.

In particular, the electronic control unit **272** on the basis of the signal received from the first distance sensor **S7** determines if/when the pair of front legs **22** are fully loaded onto (and superimposed on) the loading surface **L**, i.e. if at least the front half of the stretcher **20** is fully loaded onto (and superimposed on) the loading surface **L**.

Once the electronic control unit 272 has determined that the front half of the stretcher 20 is fully loaded onto (and superimposed on) the loading surface L, it is configured to command the lifting of the pair of rear legs 23 (by detaching them from the ground), by actuating the first rear actuator 242, up to their raised angular end stroke position.

When the pair of rear legs 23 is in its raised angular end stroke position, the electronic control unit 272 can first confirm/verify the reaching thereof by querying the first front angle sensor S2.

When the pair of rear legs 23 is raised from the ground, the load of the stretcher 20 is supported by the loading surface L and by the support coupling 32 (i.e., by the second coupling 322, which has an anti-tipping function).

Furthermore, the intermediate coupling element 313 prevents the slide 312 and the stretcher 20 loaded thereon from sliding in the direction of approach to the rear end position.

When the pair of rear legs 23 is in its raised angular end stroke position, the operator can advance the stretcher 20 (by pushing it horizontally), sliding it until the slide 312 is brought to its front end position and locked there by the coupling element 313 (and the support coupling 32 is already in its front end position).

In this position, the stretcher 20 couples, through its coupling portions 210, with the safety hooks 311 of the guide 31 (retaining the support element 32 in its front end position and the slide 312 in its front end position).

The electronic control unit 272 is configured to detect the successful and correct coupling between the coupling portions 210 and the safety hooks 311, for example by querying each second proximity sensor S13 and by receiving from it a respective signal of successful coupling.

For example, the electronic control unit 272 is configured to finish the loading sequence on the basis of a signal emitted and received by each second proximity sensor S13 and indicative of the successful coupling.

The electronic control unit 272 of the stretcher 20 is configured to perform (and/or assist in performing) a sequence of unloading the stretcher 20 from the loading surface L of the ambulance V, i.e., from the loading/unloading apparatus 30.

While performing the unloading sequence, the operator may (or must) hold down an unloading button and/or initiate an unloading sequence via the user interface 271, the release of such a button safely locks any handling of the stretcher 20.

First, the operator and/or electronic control unit 272 mechanically releases the coupling portions 210 from the safety hooks 311.

The electronic control unit 272 is configured to detect the successful and correct release between the coupling portions 210 and the safety hooks 311, for example by querying each second proximity sensor S13 and by receiving from it a respective signal of successful release.

At this point, the operator pulls the stretcher 20 axially and posteriorly with respect to the ambulance V until the slide 312 is constrained by the intermediate coupling element 313 (which locks its sliding towards the rear end position).

At this point, the electronic control unit 272 checks the position of the stretcher with respect to the loading surface L, in particular, it queries the first distance sensor S7.

In particular, the electronic control unit 272 on the basis of the signal received from the first distance sensor S7 determines if/when the pair of rear legs 23 are fully unloaded from (misaligned in plan from) the loading surface L, or if

at least the rear half of the stretcher 20 is fully unloaded from (and misaligned in plan from) the loading surface L.

Once the electronic control unit 272 has determined that the rear half of the stretcher 20 is fully unloaded from (and misaligned in plan from) the loading surface L, the same electronic control unit 272 is configured to command the lowering of the pair of rear legs 23 (until the rear wheels 231 contact the ground), by actuating the first rear actuator 242.

The electronic control unit 272 is configured to determine a correct rest on the ground of the rear wheels 231 based on a signal received from a sensor of the sensor arrangement, for example from the rear pressure sensor S15 or from the third distance sensor S12 or from the inclinometer S16.

For example, the electronic control unit 272 can be configured to detect, via the rear pressure sensor S15, a pressure value (in the hydraulic circuit of the first rear actuator 242) and compare this pressure value with a reference value thereof.

If the pressure value exceeds this reference value, the electronic control unit 272 is configured to determine the correct rest on the ground of the rear wheels 231.

Alternatively or additionally, the electronic control unit 272 may be configured to detect via the third distance sensor S12 and/or via the inclinometer S16 a change in inclination of a portion of the stretcher 20 (e.g. of the support frame 21) and determine the correct rest on the ground of the rear wheels 231 as a function or based on the detected change in inclination.

Once the correct rest on the ground of the rear wheels 231 has been determined, the electronic control unit is configured to stop the lowering of the pair of rear legs 23 (by stopping the first rear actuator 241).

At this point, the electronic control unit 272 is configured to command and actuate the unlocking arrangement, for example by commanding the second pin 267 to move to its extracted position, so as to unlock the locking arrangement, i.e. the intermediate coupling element 313 (to allow the slide 312 to slide on the fixed rail towards its rear end position).

The operator, therefore, can pull the stretcher 20 and then the slide 312 towards the rear end position and, the cam element 314 on the fixed rail 310 releases the front locking element 323 allowing the support coupling 32 to be able to move towards the rear end position.

When the support coupling 32 (and the slide 312) arrives at its rear end position and is locked therein by the front coupling element 323, the electronic control unit 272 is configured to detect this position, for example by querying the second distance sensor S8 (which detects the distance of the stretcher 20 from the ground).

At this point, the electronic control unit 272 is configured to command the lowering of the pair of front legs 22 by actuating the first front actuator 241.

The electronic control unit 272 is, for example, configured to stop the lowering of the pair of front legs 22 as a function of a signal received from the second limit switch sensor S11.

In particular, the lowering of the pair of front legs 22 (when they start to touch the ground) causes the lowering of the coupling head 260 from the upper end stroke to the lower end stroke and this lowering is indicative of (a height of the support frame 21 and/or) a load no more bearing on the coupling body 26 (i.e. on the coupling head) detected by means of the second limit switch sensor S11.

In fact, when the load bearing on the coupling head 260 is lower than a predetermined loading value, the coupling head 260 moves to its lower end stroke.

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In this position, the load of the stretcher **20** placed on the pair of front legs **22** (and pair of rear legs **23**) and frees the coupling head **26**.

In addition, the electronic control unit **272** is configured to command and actuate the unlocking arrangement, for example by commanding the first pin **265** to move to its extracted position, so as to unlock the coupling between the coupling head **260** and the coupling seat **320**, i.e. by unlocking the second coupling **322**, so as to be able to free the stretcher **20** from the loading/unloading apparatus **30** and freely move it.

The invention thus conceived is susceptible to many modifications and variants, all falling within the same inventive concept.

Moreover, all the details can be replaced by other technically equivalent elements.

In practice, any materials and also any contingent shapes and sizes may be used, depending on the needs, without departing from the scope of protection of the following claims.

The invention claimed is:

1. A stretcher, which comprises:

a support frame for supporting a patient;

a pair of front legs rotatably coupled to the support frame around a first rotation axis, wherein each front leg supports a front wheel holder frame hinged to the respective front leg around a first oscillation axis parallel to the first rotation axis and supporting a front wheel resting on a rest plane, wherein the front wheel pivots around a respective first pivot axis orthogonal to the first oscillation axis;

a pair of rear legs rotatably coupled to the support frame around a second rotation axis, wherein each rear leg supports a rear wheel holder frame hinged to the respective rear leg around a second oscillation axis parallel to the second rotation axis and supporting a rear wheel resting on the rest plane, wherein the rear wheel pivots around a respective second pivot axis orthogonal to the second oscillation axis;

a handling arrangement for the front wheel holder frame and the rear wheel holder frame configured to independently vary an inclination of the front wheel holder frame around the first oscillation axis and of the rear wheel holder frame around the second oscillation axis;

a control module operatively connected to the handling arrangement and configured to actuate the handling arrangement so as to keep the respective first pivot axis and the respective second pivot axis always orthogonal to the rest plane during handling of the stretcher, with the front wheels and the rear wheels configured to rest through rolling contact on the rest plane and with each front wheel or rear wheel pivoting in a free or controlled and/or lockable manner around the respective first pivot axis.

2. The stretcher according to claim 1, wherein the control module comprises or is associated with a sensor arrangement, including at least one first front angle sensor associated with the pair of front legs, wherein the first front angle sensor is configured to detect an angular position of the pair of front legs with respect to the support frame;

at least one first rear angle sensor associated with the pair of rear legs, wherein the first rear angle sensor is configured to detect an angular position of the pair of rear legs with respect to the support frame;

at least one second front angle sensor associated with at least one of the front wheel holder frame, wherein the second front angle sensor is configured to detect an

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angular position of the respective front wheel holder frame with respect to one of the respective front legs; and

at least one second rear angle sensor associated with at least one of the rear wheel holder frame, wherein the second rear angle sensor is configured to detect an angular position of the respective rear wheel holder frame with respect to one of the respective rear legs.

3. The stretcher according to claim 2, wherein the sensor arrangement also comprises:

at least one front absolute potentiometer, associated with and between at least one of the pair of front legs and the front wheel holder frame, and at least one rear absolute potentiometer, associated with and between at least one of the pair of rear legs and the rear wheel holder frame.

4. The stretcher according to claim 1, further comprising an actuation arrangement provided with a first front actuator, which moves the pair of front legs and which interconnects the support frame and the pair of front legs, and a first rear actuator which moves the pair of rear legs and which connects the support frame and the pair of rear legs.

5. The stretcher according to claim 4, wherein the handling arrangement comprises:

a second front actuator for each front leg of the pair of front legs, wherein each second front actuator moves the front wheel holder frame and interconnects the respective front leg of the pair of front legs and the respective front wheel holder frame; and

a second rear actuator for each rear leg of the pair of rear legs, wherein each second rear actuator moves the rear wheel holder frame and interconnects the respective rear leg of the pair of rear legs and the respective rear wheel holder frame.

6. The stretcher according to claim 2, further comprising an actuation arrangement provided with a first front actuator, which moves the pair of front legs and which interconnects the support frame and the pair of front legs, and a first rear actuator which moves the pair of rear legs and which connects the support frame and the pair of rear legs;

wherein the handling arrangement comprises:

a second front actuator for each front leg of the pair of front legs, wherein each second front actuator moves the front wheel holder frame and interconnects the respective front leg of the pair of front legs and the respective front wheel holder frame; and

a second rear actuator for each rear leg of the pair of rear legs, wherein each second rear actuator moves the rear wheel holder frame and interconnects the respective rear leg of the pair of rear legs and the respective rear wheel holder frame; and wherein the control module comprises an electronic control unit operatively connected to the sensor arrangement, to the second front actuator and to the second rear actuator, wherein the electronic control unit is configured to perform the steps of:

detecting a first front angle value by the first front angle sensor and a first rear angle value by the first rear angle sensor;

determining a first front compensation angle value and a first rear compensation angle value on a basis of the first front angle value and of the first rear angle value detected; and

commanding each second front actuator to perform a compensation rotation of the respective front wheel holder frame by the determined first front compensation angle value and/or each second rear actuator

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to perform a compensation rotation of the respective rear wheel holder frame by the determined first rear compensation angle value.

7. A stretcher, which comprises:
- a support frame for supporting a patient;
 - a pair of front legs rotatably coupled to the support frame around a first rotation axis, wherein each front leg supports a front wheel holder frame hinged to the respective front leg around a first oscillation axis parallel to the first rotation axis and supporting a front wheel resting on a rest plane, wherein the front wheel pivots around a respective first pivot axis orthogonal to the first oscillation axis;
 - a pair of rear legs rotatably coupled to the support frame around a second rotation axis, wherein each rear leg supports a rear wheel holder frame hinged to the respective rear leg around a second oscillation axis parallel to the second rotation axis and supporting a rear wheel resting on the rest plane, wherein the rear wheel pivots around a respective second pivot axis orthogonal to the second oscillation axis;
 - a handling arrangement for the front wheel holder frame and the rear wheel holder frame configured to vary, independently, an inclination of the front wheel holder frame around the first oscillation axis and of the rear wheel holder frame around the second oscillation axis;
 - a control module operatively connected to the handling arrangement and configured to actuate the handling arrangement as to keep the respective first pivot axis and the respective second pivot axis always orthogonal to the rest plane, wherein the control module comprises or is associated with a sensor arrangement, including
 - at least one first front angle sensor associated with the pair of front legs, wherein the first front angle sensor is configured to detect an angular position of the pair of front legs with respect to the support frame;
 - at least one first rear angle sensor associated with the pair of rear legs, wherein the first rear angle sensor is configured to detect an angular position of the pair of rear legs with respect to the support frame;
 - at least one second front angle sensor associated with at least one of the front wheel holder frame, wherein the second front angle sensor is configured to detect an angular position of the respective front wheel holder frame with respect to one of the respective front legs; and
 - at least one second rear angle sensor associated with at least one of the rear wheel holder frame, wherein the second rear angle sensor is configured to detect an angular position of the respective rear wheel holder frame with respect to one of the respective rear legs, an actuation arrangement provided with a first front actuator, which moves the pair of front legs and which interconnects the support frame and the pair of front legs, and a first rear actuator which moves the

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pair of rear legs and which connects the support frame and the pair of rear

wherein the handling arrangement comprises:

a second front actuator for each front leg of the pair of front legs, wherein each second front actuator moves the front wheel holder frame and interconnects the respective front leg of the pair of front legs and the respective front wheel holder frame; and

a second rear actuator for each rear leg of the pair of rear legs, wherein each second rear actuator moves the rear wheel holder frame and interconnects the respective rear leg of the pair of rear legs and the respective rear wheel holder frame; and wherein the control module comprises an electronic control unit operatively connected to the sensor arrangement, to the second front actuator and to the second rear actuator, wherein the electronic control unit is configured to perform the steps of:

detecting a first front angle value by the first front angle sensor and a first rear angle value by the first rear angle sensor;

determining a first front compensation angle value and a first rear compensation angle value on a basis of the first front angle value and of the first rear angle value detected; and

commanding each second front actuator to perform a compensation rotation of the respective front wheel holder frame by the determined first front compensation angle value and/or each second rear actuator to perform a compensation rotation of the respective rear wheel holder frame by the determined first rear compensation angle value,

wherein the determination step comprises calculating the first front compensation angle value and the second rear compensation angle value by the following formula:

$$\gamma = 90^\circ - \beta - \xi; \text{ and} \quad \text{a)}$$

$$\theta = 90^\circ - \alpha - \xi \quad \text{b)}$$

wherein γ is the first rear compensation angle, β is the first rear angle value, θ is the first front compensation angle, α is the first front angle value and ξ is an inclination angle of the support frame with respect to the rest plane.

8. The stretcher according to the claim 7, wherein the inclination angle ξ of the support frame with respect to the rest plane is calculated as a function of the first front angle value α and the first rear angle value β , the inclination angle ξ is calculated with the following formula:

$$\xi = \tan^{-1} \left[\frac{(s \cdot \cos(\beta) + s \cdot \cos(\alpha) + i) / (s \cdot \sin(\beta) - s \cdot \sin(\alpha))}{s} \right], \quad \text{c)}$$

wherein s is a length of each front leg of the pair of front legs and each rear leg of the pair of rear legs and i is a wheelbase between the first rotation axis and the second axis of rotation.

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